

Gathering Requirements for Stress Management in Autistic Adults for the Development of Augmented Reality Self-Interventions

JOHN J. MCGOWAN and IAIN P. MCGREGOR, School of Computing, Engineering, and the Built Environment, Edinburgh Napier University, Edinburgh, United Kingdom of Great Britain and Northern Ireland

Autistic adults typically encounter multiple daily stressors, particularly in workplace settings or public spaces, often resulting in sensory overload. To address these challenges, technological interventions have previously been developed to alleviate stress, enhance cognitive functioning and overall well-being. Tools such as mobile devices, virtual reality (VR), and augmented reality (AR) applications have shown promise in supporting and advancing clinical practices by utilizing sensory-based technologies. In this article, an extension of a published 2023 survey, we present the second phase of research concerning a series of semi-structured interviews with autistic participants and caregivers. We report on the identification of currently used technological preferences, and the perceived potential for new technologies. Coping mechanisms employed and reactions to stress are detailed, trigger locations and stressful scenarios are identified, as well as masking techniques used by autistic adults. The results from the study have identified interest in developing new applications based on existing technologies, with aural sensitivity identified as the main stimuli that needs managed. Participants highlighted the importance of planning as a coping mechanism and indicated that familiarity, in audio and visual feedback, helps to calm participants in times of stress. Respondents commented on existing forms of audio-visual distractors used by autistic people to cope with stressful scenarios, for example, the use of music, and stimming, which could be employed within potential applications to reduce stress. The overarching goal of this research is to address the question, “How can the potentially debilitating effects of stress in autistic adults be mitigated by a self-customizable approach to the design of accessible technologies?” The insights gained will inform future research regarding the design of technological self-interventions for stress management in autistic adults. The findings further identify practical design requirements such as adaptive audio masking, context-aware planning tools, and customizable multimodal feedback to support real-time stress regulation.

CCS Concepts: • **Human-centered computing** → **Accessibility technologies; Empirical studies in interaction design;**

Additional Key Words and Phrases: Autism, Stress Triggers, Stress Management

ACM Reference format:

John J. McGowan and Iain P. McGregor. 2025. Gathering Requirements for Stress Management in Autistic Adults for the Development of Augmented Reality Self-Interventions. *ACM Trans. Access. Comput.* 18, 4, Article 15 (December 2025), 34 pages.
<https://doi.org/10.1145/3777370>

Authors' Contact Information: John J. McGowan (corresponding author), School of Computing, Engineering, and the Built Environment, Edinburgh Napier University, Edinburgh, United Kingdom of Great Britain and Northern Ireland; e-mail: j.mcgowan@napier.ac.uk; Iain P. McGregor, School of Computing, Engineering, and the Built Environment, Edinburgh Napier University, Edinburgh, United Kingdom of Great Britain and Northern Ireland; e-mail: i.mcgregor@napier.ac.uk.



This work is licensed under Creative Commons Attribution International 4.0.

© 2025 Copyright held by the owner/author(s).

ACM 1936-7236/2025/12-ART15

<https://doi.org/10.1145/3777370>

1 Introduction

This research is concerned with the identification and management of stress among autistic people, along with the use and adoption of technologies those on the autism spectrum use in helping address stress. Qualitative data from a series of semi-structured interviews were used to enhance the understanding from the results of an initial survey [91], feeding into baseline requirements for the design and development of a proposed stress management **augmented reality (AR)** tool. Technological and non-technological self-interventions have been designed to help autistic people overcome stress-related challenges, for example, in communication, social interaction and in living independently [14, 23]. Previous studies indicated that technology demonstrated major benefits and interest for autistic people [18, 105]. Ideally, however, software designers need to raise their awareness about the variety of issues autistic people face. For example, their cultural background and level of access to technology, to help understand that developing future technology-based products requires more than mere effectiveness [74]. A 2023 review of assistive technologies for autistic adults identified a general lack of requirements-driven design involving the target audience, compounded by non-standardized evaluation processes [152]. The authors commented that few studies interviewed target users themselves, often due to the difficulties associated with autistic interviewees feeling comfortable in communicating their needs. Consequently, it is key to prioritize meeting important user goals while aligning with individual users' distinct interests and abilities.

The growth of assistive technologies for autistic individuals has seen developments in both hardware and software. Wearable devices with biosensors enable real-time monitoring of physiological indicators, for example, heart rate and eye gaze, aiming to facilitate personalized interventions to manage stress [72]. **Virtual reality (VR)** and AR platforms offer digital environments and augmentation for therapeutic interventions, such as exposure therapy [88] and social skills training [38], providing a more controlled setting for autistic individuals to cope with sensitivity issues and generalize social skills. More recently, advancements in AI have led to the development of intelligent systems capable of analyzing behavioral patterns and providing tailored feedback and support in real-time [63]. Examples include AI-driven communication aids that adapt to the user's preferences and needs, enhancing their ability to express themselves effectively. As technology continues to evolve, the availability of assistive technology for autistic individuals should ideally be offering increasingly sophisticated customized solutions, to address their diverse needs, and promote independence and inclusion, based on personal data and use. However, the underlying complexities of the technology used should be hidden from the users, as they will not require a comparable learning curve to interact with the system.

This article is an expanded version of a conference paper [91] presented at ASSETS' 23. That paper focused on results from an initial study where 202 participants, including autistic adults and caregivers of autistic people, completed an online survey. The data collected related to common triggers and sensory information recognized within autistic people, as well as from caregivers who may identify catalysts. Additionally, reflections on contemporary assistive technologies were provided, along with where potential may lie in using alternative applications, as a means of assisting high-functioning autistic adults. The article discussed the potential of technology-based self-interventions to mitigate stress among autistic adults. Examples of such interventions included wearable devices, like smartwatches, for real-time monitoring of biometric information, mobile applications offering personalized coping strategies and **extended reality (XR)** environments for immersive relaxation exercises. By designing technological solutions tailored to the specific needs and preferences of autistic adults, the study demonstrated the potential to empower individuals to manage stress more effectively and enhance their overall well-being. The current article underlines the importance of understanding stress triggers in autistic adults and highlights the promising role

of technology in developing personalized self-interventions to support stress management in this population.

2 Background and Related Work

This section discusses the background and challenges associated with **autism spectrum disorder (ASD)** and stress, as well as some of the related work detailing multimodal interventions employed to counter these challenges. In addition, effective methods of evaluating these interventions are discussed.

2.1 Autism and Stress

ASD is a complex neurodevelopmental condition characterized by a broad range of challenges in social interaction, communication, and behavior [49]. It is aptly referred to as a spectrum condition due to the variety of reactions that can vary widely across individuals. ASD encompasses a diverse group of conditions, and its prevalence has been steadily increasing, affecting approximately 1 in 36 children in the United States according to the Center for Disease Control and Prevention [84]. Many autistic people advocate for identity-first terminology concerning references to the condition [9]. Subsequently, for this article, terms related to autistic participants will be employed, for example, referring to autistic people or autistic participants.

Autistic people may be challenged with understanding and using verbal and non-verbal communication, interpreting social cues, and engaging in reciprocal relationships [81, 99, 109]. This can lead to challenges in forming and maintaining friendships, as well as understanding body language and the unspoken rules of social interaction. Camouflaging is a common mechanism for coping, which can often result in negative mental health consequences, as can attempting to “pass as normal” or adaptive morphing [24, 34]. Sensory sensitivities are common among autistic individuals, who may experience heightened or diminished responses to sensory stimuli such as unpredictable sounds, certain types of lighting, undesired textures, and unwanted smells [124]. Sensory challenges can impact daily functioning, making certain environments overwhelming or uncomfortable. For example, in the workplace, human and mechanical sounds, as well as strip lighting, can be considered stressors, or on public transport, where olfactory, temperature, and touch stimulus can cause additional problems [151]. Subsequently, this can lead to high levels of stress in different scenarios, as over-stimulation can negatively affect cognitive functioning [83].

Repetitive behaviors and a preference for routine can be characteristic for autistic people as a coping mechanism, providing a sense of predictability and agency [138]. Autistic individuals may also engage in self-stimulating behavior (stimming), have intense interests in specific topics, and resist change in their daily routines as alternative ways to manage sensory overload [12]. One prominent challenge autistic individuals face is the co-occurrence of comorbid conditions, including intellectual disabilities, epilepsy, **attention-deficit/hyperactivity disorder (ADHD)**, anxiety, and depression [29]. Addressing these are important for providing comprehensive care and support. In dealing with sensory challenges, autistic adults may use masking techniques within social situations, as a means of disguising their condition, which can cause stress and lead to further health issues [61].

Access to appropriate educational and employment opportunities can be taxing for autistic people, as well as the working environments that neurotypical people do not find problematic [151]. For example, open-plan offices with fluorescent lighting can present challenges to neurodiverse workers, while not affecting neurotypical employees, as unwanted sound and light can be overwhelming stressors [55]. Subsequently, tailoring educational strategies to meet autistic individuals’ unique learning styles and providing workplace accommodations can significantly enhance success in academic and professional settings [161]. A recent study by Ara et al. [8] highlighted some of

the same external and social stimuli that can be challenging for autistic workers and job seekers and suggests simple adjustments can be made to cater for neurodiverse workers quite easily. However, the authors also suggest the need for greater training to address stigma surrounding autism. Throughout their lifespan, autistic individuals and their families can encounter societal misconceptions, stigma, and a lack of understanding, which contributes to social isolation, negative health conditions, and hinders access to support services [42]. Advocacy for autism awareness and acceptance is essential to foster a more inclusive and supportive society [25], and as such, is one of the overarching aims of this research.

2.2 Related Work—Current Interventions

Current interventions for alleviating stress in autistic individuals include both technological and non-technological approaches. By integrating these approaches, comprehensive support systems can be tailored to individual needs, beyond promoting well-being and adaptive functioning in autistic individuals.

2.2.1 Non-Technological. Non-technological strategies include **cognitive-behavioral therapy (CBT)**, which has shown to be effective in treating acute stress and supported by empirical evidence [28]; however, other studies have also demonstrated a variety of efficacy for autistic people and that more evidence is required [153]. Other cognitive therapies include mindfulness practices, and sensory integration therapy, which aim to modify thought patterns, promote relaxation, and improve sensory processing [1]. Meditation has also been seen to be a useful activity to reduce stress for autistic people, in a bid to overcome sensory overload and focus on directing and determining which senses to focus on [130]. In addition, the use of music, both listening to and participation, is well-established as an intervention that has shown significant potential in improving the oral-motor activities of autistic people, as well as having a positive impact upon social communication, motor skills, and reducing stress levels [21, 58]. Studies that use yoga and mindfulness-based programs as supplemental interventions, for autistic children and adolescents, suggest efficacy in a number of areas. Although few studies have reported improvements in the core symptoms of autism, they may improve a variety of prosocial behaviors, including communication and imitative behaviors; increased tolerance of sitting and of adult proximity; self-control; and social communication [131]. For example, Radhakrishna [117] employed an Integrated Approach to Yoga Therapy where the authors reported improvements in communication, imitative behaviors, and motor control. Singh et al. [134] conducted studies using a mindfulness technique known as “Meditation on the Soles of the Feet” to help manage aggressive impulses of youth with Asperger Syndrome. Findings indicated that all three adolescents reduced instances of their aggressive behaviors from a mean of 14–20 per week during baseline, to 4–6 per week during mindfulness training, subsequently to zero rates during the last 4 weeks of intervention. Additionally, reductions in lethargy, social withdrawal, and non-compliance, were observed [22]. However, the need for additional evidence-based research is key in supporting these findings.

Garcia and Dattilo [52] studied the effects of a leisure program on autistic adults over the course of a year (for example, interacting with media, engaging in exercise, playing games, attending events, and participating in other social activities) and found a significant increase in the four factors of quality of life that were measured. These included satisfaction, independence, competence, and social interaction, as well as the total score for quality of life from baseline to the end of the intervention 12 months later. Leisure activities that do not require high skill levels and promote team environments have been recommended for autistic individuals, for example, rock climbing. In a small-scale study, Oriel et al. [106] reported no significant change in attention test scores, but the

climbing activities demonstrated an increase in self-esteem from adolescent autistic participants, as well as gaining positive feedback from parents regarding social skills.

Another form of common non-technological (and technological—see Section 2.2.2) intervention includes self-stimulating behavior (stimming), such as hand flapping and body rocking, which can be seen as a self-regulatory mechanism for autistic people [111]. From a neurotypical perspective, stimming still lacks social acceptance and understanding [66], although attempts are being made through media to inform public opinion [2, 15]. However, stimming is subject to a very different kind of interpretation in neurodiverse people, in which such behaviors are embraced as productive, communicative, pleasurable, and even socially valuable for those who perform them [12]. Additionally, stim toys are becoming more commonplace, commercial, and accepted as a means of positive distraction, to help focus, or to develop motor and perceptive skills for a neurodiverse population [132]. For example, *focus toys* provide a positive distractor via joining together interlocking pieces, cubes or fidget spinners; *squishy* and *tactile play* toys create a tactile sensory feedback mechanism designed to reduce stress [146].

2.2.2 Technological. In addressing sensitivity and stress in autism, technology can play an important role. One common approach involves using headphones to listen to music as a positive distractor [133] and to employ noise cancelation features to address auditory sensitivity [62]. Headphones help individuals regulate auditory stimuli by listening to their preferred music or white noise while reducing external noises as an aid to managing external auditory stimuli. Sunglasses are also used to mitigate sensitivity to different forms of lighting, as they can filter and reduce the intensity of visual stimuli, enhancing comfort and reducing distress [100].

Stim toys are items or props sought by the participants or provided by carers. Less formal than a therapeutic intervention, they are designed for autistic children and adults for sensory regulation [48]. They offer tactile stimulation for self-soothing and redirecting attention from overwhelming sensory inputs [31]. Mobile technologies provide access to specialized apps designed for individuals with autism [139]. These apps offer a variety of functions, including sensory regulation exercises, visual schedules, and communication aids [100]. Mobile technologies aim to allow autistic individuals to manage their sensory experiences and navigate social or working contexts independently [149].

Wearable technologies, equipped with biosensors and AI algorithms, can monitor physiological indicators of stress in real-time [75]. They can provide personalized biometric feedback, for example, heartrate and related individual arousal levels. Some applications offer relaxation techniques or sensory grounding exercises through mobile technologies, including XR platforms [29]. Developing XR technologies demonstrate potential for creating immersive environments for desensitization therapy and social skills training, for example, providing relaxation experiences, such as virtual nature scenes or calming activities which promote relaxation and emotional regulation [26, 143]. However, care must be taken with XR use not to isolate target users from the real world as concerns exist regarding the introduction of tools that may encourage obsessive behaviors or social rigidity, in particular, via VR headsets which may further contribute to autistic individuals' avoidance of social situations and lead to isolation from their peers [89, 101, 141]. By integrating diverse technological modalities, support systems can be established to promote the well-being and adaptive functioning of individuals with autism in stressful situations.

2.3 Evaluation of Current Interventions

Quantitative and qualitative methods have been employed to evaluate the effectiveness of current interventions. Quantitative assessments include self-report questionnaires [67], caregiver reports [102], or clinician-administered assessments to quantify changes in stress levels, anxiety symptoms,

or related behaviors over time [135]. Additionally, physiological measures such as heart rate variability, skin conductance, and cortisol levels may be utilized to provide objective indicators of stress and anxiety levels before and after intervention implementation [40].

Qualitative evaluation methods have also been frequently employed to gather in-depth insights into the experiences and perspectives of autistic individuals and their caregivers regarding intervention effectiveness. Qualitative approaches such as interviews, focus groups, and participant observation allow for a nuanced understanding of how interventions impact various aspects of daily life, including coping strategies, social functioning, and overall well-being [36, 47]. Alternatively, mixed-methods research designs, which combine both quantitative and qualitative data collection and analysis techniques, offer a comprehensive approach to evaluating intervention outcomes through triangulating findings from multiple sources and perspectives [154].

Other methods include observational studies, conducted in naturalistic settings which may provide valuable information about the real-world effectiveness of interventions in everyday contexts [30]. Observational methods involve systematically documenting behavior and interactions before, during, and after intervention implementation to assess changes in stress and anxiety-related behaviors in naturalistic settings such as homes, schools, or community settings. By integrating multiple evaluation methods, researchers and practitioners can gain a holistic understanding of the impact of interventions on stress and anxiety management in autistic individuals, informing the development of evidence-based practices and interventions tailored to individual needs.

This work complements prior studies on technological stress interventions for autistic adults, including Robertson and Simmons [122] on sensory preferences, Fletcher-Watson et al. [50] on participatory research approaches, and Kalantari et al. [65] who provide relevant evidence on stress and emotion regulation using digital tools and wearables. Compared with these, the present study extends to adult self-intervention contexts and draws on direct end-user interviews to refine design requirements. These studies help position the current findings within emerging accessible-technology design frameworks.

3 Method

Online semi-structured video interviews were carried out over the course of 2023 with autistic participants, as well as caregivers, to gather information from alternative perspectives. The caregiver group comprised parents of autistic adults and professional carers. The broad aim of the study was seen as an extension of the first phase of this research [91]. The initial phase sought to further investigate the identification of stress triggers within high-functioning autistic adults, ways to cope with stress, as well as past, current, and potential future uses of technology that may provide the potential for new applications as a means of assisting managing stress experienced by autistic adults.

3.1 Participants

Twenty respondents took part in the study, with 15 autistic participants and five caregivers. Of the autistic participants, 44% identified as male, with the remaining 56% identifying as female. All autistic participants reported having a formal autism diagnosis, with caregivers providing information to confirm a formal diagnosis for the people under their care. Of the autistic participants, 63% reported ASD, with the remaining 37% identifying Asperger Syndrome. Asperger's was once considered as different from autism; however, in 2013 the American Psychiatric Association decided to merge the subtypes of pervasive developmental disorders into a single category of autism on the assumption that they cannot be reliably differentiated from one another [4]. However, many practices still use the older form of categorization. Seventy five percent of the autistic participants also reported having comorbid conditions, including ADHD (38%), post-traumatic stress disorder (19%), with

Table 1. Overview of Interview Questions

Section	Description
General background Questions	Related to demographics, living and working arrangements, general difficulties, and challenges.
Identifying and coping with Stress	Concerning levels of comfort and discomfort in specific locations, as well as physical aspects of stress and coping mechanisms.
Past use of technology	Preferences of technologies previously used and reasons for their rejection.
Current use of technology	Preferences of technologies currently used and their levels of support.
Interest in new technology	Potential for new technologies to assist in stress management.

additional conditions including dyspraxia, arthritis, vertigo, hypopituitarism, fibromyalgia, eating disorder, social anxiety, depression, dyslexia, developmental coordination disorder, and epilepsy. The ages of the autistic participants ranged from 19 to 63, with a mean of 36, while caregivers ranged from 45 to 58, with a mean of 51.

The University's ethical approval process prevented individuals under 18 or over 70 taking part. Participants were recruited through Autism-related organizations including Autistica [11] and Differabled Scotland [39], as well as via social media platforms such as LinkedIn [80] and X (formerly known as Twitter) [148], among others. Recruitment information highlighted that participants were required to have a formal autism diagnosis or be a carer for someone who has a formal diagnosis and be aged between 18 and 70 years. As part of the ethical procedure, only participants who were capable of verbalization were included as part of the study in accordance with university guidelines. No financial inducement was made to take part in the study.

3.2 Materials

Interviews were conducted online using Microsoft Teams [95] which was also utilized to record the conversations. Interviewees were provided with a digital copy of a participant information sheet, consent form, and the questions prior to the interview (see Supplementary Material B). Table 1 offers a summary of these sections.

3.3 Design

A semi-structured interview method was selected, which allowed participants to elaborate on their responses. This approach facilitated the emergence of additional questions through the dialogue between the interviewer and interviewee. Given the diverse manifestations of autism-related behaviors across communication, social interaction, and behavioral domains, direct answers and anecdotal insights were valuable in understanding the choices made by participants. Prior research has demonstrated the effectiveness of semi-structured interviews in gathering qualitative background information from both caregivers and from autistic adults, including the routines and preferences of each participant [41, 46].

The interview comprised 34 questions categorized into five sections (Supplementary Material B). The initial section consisted of nine questions focusing on general background and demographics. Six questions in part 2 related to identifying and coping with stress. Sections 3–5 addressed past, current, and potential interest in new technologies, respectively. The interview question design was based on the findings of the initial survey where identified themes were used as the basis to create related areas of inquiry [91]. Table 2 outlines the themes raised and the related question areas. Please refer to Supplementary Material B for the full list of questions.

Table 2. Themes and Associated Question Areas

Themes	Questions areas that address themes
Main sensitivity triggers—unwanted sounds and lighting	Identifying and coping with stress
Unfamiliarity as a stressor	
Stimming as a coping mechanism	
Usability and accessibility	Past use of technology
Use of music	Identifying and coping with stress/current use of technologies
Positive distractors	Identifying and coping with stress/current use of technologies/interest in new technologies
Ubiquitous smartphone use	Current use of technologies/interest in new technologies
Interest in VR and AR technologies	
Difficulties in identifying stress levels	Interest in new technologies
Potential solutions—use of headphones, smartphones, and AR	

For the present study, a co-created coding process was carried out between the authors using NVivo [116], applying a qualitative grounded-theory–inspired thematic analysis [150] to interpret interview transcriptions. Quotations were tagged with codes in an open manner, guided by the interpretation of the content rather than predefined categories. Initially, 36 codes were identified, and subsequently condensed into 15 main group codes. The findings were organized into 15 sections, with each section corresponding to a specific transcript code (refer to Table 3 in Section 4). Quotations, phrases, sentences, and short paragraphs were identified and linked to the relevant codes based on their relevance and level of detail in addressing interview questions. The frequency of each code’s occurrence was recorded and ranked for transparency. Frequency counts were reported descriptively to indicate salience across participants, not as a proxy for thematic importance.

Regarding the suitability of the sample size for using a grounded-theory type approach, it should be noted that in related studies involving specialist groups, Creswell and Poth [35] have proposed a minimum of 20–30 participants in the grounded-theory research design, while Guest et al. [56] suggest a sample size of 12 is sufficient before data saturation and variability occurs in larger sample sizes.

3.4 Procedure

Ethical approval was granted by the University’s ethics committee, and participant consent, along with a participant information sheet and list of questions, were e-mailed to participants to ensure approval prior to interviews taking place. The interviews were conducted via Microsoft Teams, where the audio was recorded, and later transcribed into Microsoft Word documents, which were anonymized. Coding was conducted using the transcriptions from the interviews based on a qualitative grounded-theory–inspired thematic analysis [150]. Codes were allocated to quotations using an open approach where interpretation of quotations suggested appropriate codes, as opposed to predefining codes and assigning quotations to them. Following a first pass, which determined a set of 36 codes, a further pass merged the codes into a condensed set of 15, which were checked against the transcripts again for assuredness. The first pass identified general themes from the transcript, while the second pass noted similarities and combined several of the codes together under the new headings as seen in Table 2. With each pass, the authors had a discussion, looked at samples, and refined each of them until both in agreement. The agreed codes were reviewed by both authors and are described below, followed by detailed findings from the data. Coding disagreements

were resolved through negotiated consensus between the authors, and saturation was confirmed once no new subthemes emerged across the final five transcripts.

4 Results

The results are divided into 15 sections using the second pass transcript codes as headings, which include: technology, coping mechanisms, reactions, locations, assistance, stressful scenarios, comorbidity, interests, masking, friends and family, employment, opportunities, preconceptions, skills, and empathy. Interviewees commented most on technologies used (with the highest number of instances at 395) in the past, currently, as well as the potential for new technology. Coping mechanisms, with 213 instances, were the second most commented subject, referring to different actions taken by autistic people to regulate their emotions. Reactions (191 instances) refer to immediate and delayed reactions to stress which includes hypersensitivity and hyposensitivity, as well as reference to physical reactions. Due to word count restrictions, the entire dataset gathered from the interviews will not be presented here. The present study will focus on *Technology*, *Coping Mechanisms*, *Reactions*, *Locations*, *Stressful Scenarios*, and *Masking*, as the primary codes for assisting in the development of potential technologies regarding stress management. The remaining codes will be presented in future publications. Please refer to Table 3 for the outstanding instances and descriptions for each of the main, and remaining, codes.

The interviewees are referred to within the results as PA1, PA2, and PA3, representing the autistic participants, and C01, C02, and so on, representing carers, for the purposes of anonymization. In addition, any references within transcripts to named individuals will be replaced with an X. Detailed results from each code are presented below, divided into two sections: first, from participants who have been diagnosed as autistic; and second, from carers of diagnosed autistic individuals. The distribution of codes reflects recurrent patterns rather than statistical weighting, supporting transparency in how participant priorities emerged.

4.1 Autistic Participants

4.1.1 Technology. *Technology* encompasses five sub-codes which are expanded here, including *Technology*, *Current Use of Technology*, *Current Apps*, *Potential for New Technology*, and *Reminder System*. Autistic participants were asked about any form of device that could be used for their own benefit. They commented on currently used hardware and apps, with potential for exploring familiarity, positive distractors, and reminders, within a new accessible application. The results are summarized in Tables 4 and 5, with expanded results available in Supplementary Material D.

Participants commented on the *Potential for New Technology* and is presented in Table 5, which has been further divided into themes for ease of understanding.

4.1.2 Coping Mechanisms. *Coping mechanisms* encompass seven sub-codes including *coping mechanisms*, *emotional regulation*, *focus*, *planning*, *and the agency*, *use of music*, *progress made*, and *stimming*. Autistic participants were asked about any actions they performed to assist in emotional stability. Interviewees discussed alternative ways they cope with sensory stimuli through recreational activities, peer support, the use of music, stimming, creativity, and other distractors, as well as the importance of planning and agency. The results are summarized in Table 6, with expanded results in Supplementary Material D.

4.1.3 Reactions. *Reactions* encompass five sub-codes including *reactions*, *lack of body awareness*, *physical reactions*, *sensitivity*, and *tactility*. Autistic participants discuss immediate and delayed reactions related to stress. Interviewees commented on hypersensitivity and hyposensitivity, emotional and physical reactions to stress, as well as difficulties in recognizing associated physical

Table 3. Codes, Descriptions, Percentage of Responses for Each Code (PA Indicating Autistic Participants, C Indicating Carers), Ordered by Total Instances of Each

Code	Descriptions	Total #	Participants % (PA/C)
Technology	Any form of device that can be used for the benefit of autistic individuals.	395	100
Coping Mechanisms	Actions taken by the autistic person performed to assist in emotional stability.	213	90 (87/13)
Reactions	Immediate and delayed reactions related to stress.	191	100
Locations	Physical spaces visited or inhabited by autistic people.	95	100
Stressful Scenarios	Situations that can occur that cause stress for autistic people.	89	95 (80/20)
Masking	An altered behavior, either due to social expectations or a desire to fit in. This can extend toward issues in self-trust and trust in others.	34	55 (74/26)
Assistance	Help offered or received from or by others.	91	100
Co-Morbidity	Additional conditions that the autistic participant regularly copes with.	53	90 (89/11)
Interests	Typically, strong obsessive attention to detail to provide emotional support.	50	85 (70/15)
Friends and Family	Current living scenarios of autistic participants.	24	80 (55/25)
Employment	Details regarding autistic participants' current employment.	23	65 (55/10)
Opportunities	How lifestyle, working, education, and support situations are presented to, and affect, autistic people.	15	35 (20/15)
Preconceptions	Expectations of neurotypical people from the perspective of, and concerning, autistic individuals.	13	35 (25/10)
Skills	Abilities that have been accrued over time through repetition or natural talent.	13	45 (40/5)
Empathy	Awareness of others' emotions and its effect on the autistic person.	8	20 (15/5)

The top six codes highlighted in yellow will be expanded within the results section.

changes. The results are summarized in Table 7, with expanded results available in Supplementary Material D.

4.1.4 Locations. *Locations* encompass four sub-codes including *locations*, *familiarity*, *secure space*, and *stressful locations*. Autistic participants discuss physical spaces they have visited or inhabit. They commented on familiarity being a positive major theme regarding physical spaces, digital spaces, and people, the need for secure spaces, as well as unpredictable sensory stimuli being a stress trigger. The results are summarized in Table 8, with expanded results available in Supplementary Material D.

4.1.5 Stressful Scenarios. *Stressful Scenarios* encompass two sub-codes including *stressful scenarios* and *unpredictability*. Autistic participants discuss situations that can occur that cause them stress. The interviewees commented on stress caused by unpredictable sensory stimuli which

Table 4. Overview of Technology, Current Use of Technology, Current Apps, and Reminder System

Sub-Code	Overview of <i>Technology Results</i>
Technology	Desire to buy new technology decreased as smartphones subsumed functionality of differing devices. Interviewees expressed an interest in VR and AR technologies for entertaining and therapeutic use but were wary of issues regarding nausea and being overly isolated. Need for simplicity and usability regarding making technology accessible, particularly under duress. Gaming has been a popular choice for some participants, and subsequently informed their choice of technology, for example, gaming consoles, Nintendo Wii and switch.
Current use of technology	Most commonly, connected devices like smartphones and laptops are in regular use. Some autistic participants use headphones to regulate unpredictable noise in open and public spaces, but also to listen to music. Stim toys are used as coping mechanisms for autistic participants, like spinners and fidget toys. Wearable devices like smartwatches and Garmin's are used for recreational purposes but also as reminders and for tracking physical metrics like heartrate. AR has been used as a gamified means to keep fit, which also included social communication benefits. Participants use customization in technology wherever possible but also recognize the need to limit use for their own well-being and agency, to limit obsessive and unhealthy behaviors.
Current apps	Autistic participants reported using apps that aim to help identify and address personal moods, including CBT-based apps like Youper, Daylio, and Autistic Bliss. The usability of some apps has restricted their use and adoption, for example, Brain In Hand, which was suggested as difficult to set up. Interviewees indicated that many apps used are games and employed as positive distractors and coping mechanisms. For example, Falling Sand game, math games, Tetris, puzzle games, word games, and idle clickers. However, multiplayer games did not suit all respondents and could generate negative emotions. Customization of apps is key for some participants, to help manage sensory processing, for example, via color and volume. Meditation and relaxation apps are a popular choice for some of the respondents, including Headspace, Insight Timer, Atom, and Calm. Participants noted the benefit of using synthesized and natural sounds to reduce stress within an app. Visual stimuli, including simple animations, helped augment and expedite meditative processes. Music apps including Spotify, Audible, and YouTube were a popular choice for participants as a regular means of emotional regulation. Respondents use calendars as reminders and organizational tools. Participants also use apps for other healthcare purposes, including monitoring physical status, or to perform specific muscular exercises.
Reminder system	A potential feature suggested by autistic participants, focused on the idea of having a reminder system triggered by biological metrics, that could alert the user of unnoticed physiological changes, before it became problematic. This may include heart rate, but also hydration and blood oxygen levels for health reasons. Other respondents mention the potential for simple and day-to-day reminder systems that should be fun and could also build in recuperation time for more stressful activities.

affected work and home life, the difficulties of social situations and confrontation, and changes in routine. The results are summarized in Table 9, with expanded results available in Supplementary Material D.

4.1.6 Masking. *Masking* encompasses two sub-codes including *masking* and *visibility*. Autistic participants commented on an altered behavior, either due to social expectations or a desire to fit in. This can extend toward issues in self-trust and trust in others. Participants commented on health problems related to masking, self-discipline in maintaining appearances, the use of distraction techniques, as well as discomfort felt in relation to being observed by others. The results are summarized in Table 10, with expanded results available in Supplementary Material D.

Table 5. Overview of Potential for New Technology

Themes	Overview of Potential for New Technology Results
Limitations	Limited economic reasons were suggested by some as reasons not to invest in new technology. Some participants had concerns about being too isolated, drawing attention to themselves, and using visual head-mounted technology in general, due to blocking off senses, and the weight.
Interest in AR	Others expressed an interest in AR technology and its ability to add digital information onto the existing world, for example, reminders, shopping lists, and personal biometric feedback. Participants commented that the development of new glasses-based technology designed for a wide user group could also remove some of the stigma surrounding previous discrimination. Participants suggested that technology could be used to act as an aid in difficult situations where prompts could be provided, for example, booking appointments or checking transport timetables.
Audio	Audio was suggested as a simple way to notify participants of potential issues, through simple beeps or familiar voices. More specific audio masking or noise reduction techniques were also suggested as being potentially useful to reduce unpredictable and painful noise. The importance of using audio as a positive distractor, through music, or customized natural sounds, like crunching leaves, waves, thunderstorms, or rain.
Visualization	There was also interest in using visualization of sound for participants with an audiovisual orientation. Participants were interested in the idea of gamified apps, if it provided a level of focus and well-being. Respondents suggested minimization of visual stimuli could be valuable in controlling potential overload, while others suggested introducing images into their field of vision that they would like to see. Participants noted that the lack of ability to visualize things in their mind can be problematic, and technology that could help with this would be beneficial.
Haptics	Respondents suggested implementing some kind of haptic system, like an arm band or brace, that could subtly let the user know when their stress levels are rising.
Connected-ness	Suggestions were made for an application that could connect a distressed autistic person with someone who could understand the situation and aid them. Interviewees indicated that new self-intervention technology could help people with depression more effectively than current group-based solutions in allowing them to participate in online gamified solutions.
Customization and accessibility	Customization was considered an important aspect regarding the creation of personal themes and the use of color to control mood and enhance well-being. Other forms of customization included comments on avoidance of technology as well, to reduce cognitive stress, for example, turning off all technology after a specific period. Customization was also mentioned regarding adapting to certain locations or scenarios, for example, based on the age of the user, their location, their current state of mind, and level of stress. Interviewees indicated that the usability of any new application would by necessity have to be accessible to all, including preferred use of imagery over text, but with supporting textual and audio information to aid in its understanding. However, safety measures would have to be built in for vital information like traffic and emergency services.

4.2 Carers of Autistic Individuals

4.2.1 Technology. *Technology* encompasses five sub-codes which are expanded here, including *Technology*, *Current Use of Technology*, *Current Apps*, *Potential for New Technology*, and *Reminder System*. Carers were asked about any form of device that has or could be used for the benefit of the people in their care. Participants commented on currently used hardware and apps, with potential for exploring familiarity, positive distractors, and reminders, within a new accessible application. The results are summarized in Table 11, with expanded results available in Supplementary Material C.

4.2.2 Coping Mechanisms. *Coping mechanisms* encompass seven sub-codes including *coping mechanisms*, *emotional regulation*, *focus*, *planning and the agency*, *use of music*, as well as *progress*

Table 6. Overview of Coping Mechanisms, Emotional Regulation, Focus, Planning, and the Agency, Use of Music, Progress Made, and Stimming

Sub-Code	Overview of <i>Coping Mechanisms</i> Results
Coping Mechanisms	Centering, meditative distraction to reduce stress: breathing techniques, standing on one leg, counting toes. Stimming to provide a positive distraction. Yoga and meditation were also mentioned as common techniques. Mindfulness exercises and cognitive therapies were also suggested as beneficial coping mechanisms. Recreational activities, indoor, but more commonly in open environments, are used as other coping mechanisms, for example, walking around, sports, archery, exercising, and being in nature. Regarding audio sensitivity, participants commonly use headphones to minimize sensory overload. Use of medication to control stress, however, this was not seen as generally desirable. A level of control and creativity is important as a coping mechanism, which can be expressed through games, color coding of clothes, and indoor lighting conditions. For visual sensitivity, the use of sunglasses in scenarios like driving, shopping, or being exposed to daylight. Need to shut down and reset if they are feeling overwhelmed, through short naps, watching TV. Support groups and social contact are another means of coping and socializing with other autistic people. In potentially awkward social situations, participants find an occupying task to minimize social contact. Pets can be therapeutic and act as a comfort and a distraction.
Emotional regulation	Lack of perspective at times in regulating their emotions, which can lead to being overwhelmed, and an inability to understand their own actions as an unconscious means of moderating them, like stimming. Literature helped understanding of managing well-being, "Looking After Your Autistic Self," by Niamh Garvey.
Focus	In difficult situations, where stress can dominate awareness, shifting focus to other things provides some relief. Creative activities like craftwork and using preferred tactile materials. Need for visual interest to distract themselves from emotionally challenging situations and indicated that they find difficulties in functioning properly without visual distraction, like having a view through a window. An interviewee suggested working in a group can be a better way to focus on activities, compared to solo.
Planning and agency	Having prepared responses for several scenarios, for example, social interactions, or health related visits. Respondents commented on the reliance of task lists in providing a sense of agency and in managing mood. Planning takes into consideration when there will be less people around, and subsequently less unpredictable stimuli, for example, via transportation routes, or to avoid the end of the school day. Planning for reliance on technology and its limitations, including back up batteries for mobile technology.
Use of music	Participants commented on listening to and creating music to ally and resolve their physiological and mental state in difficult emotional situations, for example, bereavements. Creating playlists for different situations, moods, for stimming, and as an emotional outlet. Three different ways that they use music as a positive distractor, a motivator, and a calming influence. Instrumental music, or music in a different language to the listeners own, can also be used to help maintain focus, as they indicate that recognizable voices can be a distraction to the task at hand. One suggested that ambient music acts as a good way to mask unwanted sounds and to allow focus on other activities. Interviewees indicated that music could have a relational effect and be used to trigger or remember certain emotions. Additionally, it was suggested that music can be used as a vehicle to share emotional states with others, indicating that for autistic people who may feel isolated or excluded, creating music can provide a level of self-esteem, as well as bringing with it a sense of community.
Progress made	Participants suggested that by managing stress and emotions more effectively, they have been able to perform organizational work tasks that would not have been possible before. Change of perspective following diagnosis, evaluated position as a ND person masking within a NT world. By forcing themselves out of their comfort zone, they could achieve more, academically, and socially.
Stimming	In the past, autistic participants would use a live music experience to express themselves through stimming. P01 commented on being reprimanded for involuntary hand motion as a child but suggested that by using their hands in a productive way, the actions could be personally beneficial and act as a natural form of masking. Others suggested that being more open about stimming was something that should be embraced and that others would understand and accept this as a healthy autistic trait. They also suggested that other activities could be experienced as a form of stimming, including driving.

made and *stimming*; however, carers did not comment on the last two sub-codes here. Carers were asked about actions taken by the autistic person they care for, performed to assist in emotional stability. Interviewees discussed the alternative ways autistic people cope with sensory stimuli through recreational forms, family support, the use of music and other distractors, as well as the

Table 7. Overview of Reactions, Lack of Body Awareness, Physical Reactions, Sensitivity, and Tactility

Sub-Code	Overview of Reactions Results
Reactions	Stress can create physical pain in different areas of the body, including headaches, back pain, or joint pain. Stress can manifest physically via the skin or as a negative emotional reaction. Fatigue and lengthy periods of recovery, sometimes lasting for days or months, depending on stress severity. Respondents suggested activities that did not feel natural to them were more likely to cause fatigue, for example, presenting in front of other people. Travelling could be debilitating due to sensory overload and levels of unpredictability. In a similar manner, participants commented on stress-related panic when shopping in public. Participants noted that the buildup of stress-related incidents could have serious long-term health-related effects, including breakdowns, alcoholism, or addiction problems. Accumulated effects of frustration and lacking self-confidence, due to inability to fully express themselves. Too much introspection can have adverse effects, including depression. Physical effects of using VR included disorientation and nausea. Stressful health-related appointment, the mind shuts down the body as a means of coping with the situation. Frustrations, including relationship issues and malfunctioning technology, led to destructive reactions. Others commented on unwanted touch having an unwanted lasting effect, both physically and cognitively.
Lack of body awareness	Autistic people find it difficult to monitor and maintain their physical well-being, and they commented on an inability to recognize physical changes. This can include discomfort due to unnoticed changes in temperature. Participants indicated that increased amounts of stress can lead to dissociation and memory loss. Interviewees suggested that a level of hyposensitivity, combined with avoidance of challenging situations, can lead to delayed physical and emotional impact later in life, typically manifesting as ill health. Physiological changes in their body have immediate impact on emotional state can lead to a lack of self-control.
Physical reactions	Aware of physical consequences to their planned actions, such as fatigue. Stress recognized through different phases, heightened respiratory function, pain in different body parts. Other comments also mention headaches and chest pain, as well as stomach pain and loss of appetite. Cognitive processing and other conditions, including irritable bowel syndrome noted as reactions to stress. Previously held thoughts about anxiety/emotional outbursts could be misconstrued sensory process reactions. Physical pain due to noisy, stressful auditory environments: unpredictable weather, people, or animals. Use of certain lighting types that caused headaches and nausea: strip lights in offices, doctors, or dentists. Conceptual idea of their senses projected as a vulnerable bubble surrounding them. Rising sense of being overwhelmed made participants feel as though they were disconnected from their body. Participants noted that feeling overwhelmed with sensory stimuli can affect speech in a detrimental fashion. Immediate and extended periods of fatigue a common reaction to stress and recovery.
Sensitivity	Anxiety levels affected by aural sensitivity: mechanical hums, and by unpredictable noises made by people. They also pointed out that small unidentified sounds can be a major distraction. Difficulties associated with public transport and stations regarding noise. Irony of being a loud person sensitive to sound, usually due to the unpredictable nature of people. Some indicated that background noise can be a major distraction in their ability to focus. Affected negatively by bright light, headaches, and nausea, daylight, and artificial light (LED/strip lighting). Customize screen-based technologies using preferred color or dimming brightness to control stress levels. Olfactory sensitivity, certain stimuli can overwhelm, for example, sushi, glue, or smells created by people. Memories of bad sensory experiences in the past can affect present cognitive functioning in the present. Hyposensitive, for example, not feeling physical pain, or not immediately feeling the effects of photosensitivity. Sensitivity can affect working choices and restricting environments like the classroom, or brightly lit offices.
Tactility	Sensitivity of materials on skin and difficulties in coping with them. Extends to unwanted contact from people. Can affect choices regarding eyesight: not touching eyeball to put in contact lenses or glasses being too heavy. Feel of certain materials can affect their choice of activity in both positive and negative ways. Unwanted touch from people more stressful than from objects or materials, typically in crowded public spaces.

importance of planning and agency. The results are summarized in Table 12, with expanded results available in Supplementary Material C.

4.2.3 Reactions. *Reactions* encompass five sub-codes including *reactions*, *lack of body awareness*, *physical reactions*, *sensitivity*, and *tactility*. Carers discuss immediate and delayed reactions related to stress, for the autistic people in their care. Interviewees commented on sensitivity, emotional, and physical reactions to stress, as well as difficulties in recognizing associated physical changes. The results are summarized in Table 13, with expanded results available in Supplementary Material C.

Table 8. Overview of Locations, Familiarity, Secure Space, and Stressful Locations

Sub-Code	Overview of <i>Locations</i> Results
Locations	Locations where they feel comfortable, including being in nature, home, as well as having time alone. Other respondents appreciated the quiet of alternative locations like graveyards and woodlands. Unwanted noise plays a part in choosing comfortable locations, natural sounds help to reduce stress. Noisy public gatherings can be accessible with appropriate facilities, like quiet zones for people to recover. Familiarity as a main factor in feeling comfortable in locations, whether it be people or places of work. Visual stimuli can affect comfortable locations, some prefer minimal approach, other colors and lighting.
Familiarity	Participants suggested a level of control and predictability were important factors, including knowing where the exits are, choosing preferred seating in cafes, and being at home in their own comfort zone. Respondents implied that communicating with other autistic people had a positive effect. Participants commented on familiar sounds being calming stimuli, for example, familial voices. Learning routines as a positive way to relax, for example, in learning a craft or musical instrument.
Secure space	Physical spaces to go in order to cope with being overwhelmed with stimuli, typically at home. They suggest workplaces can be made better with protected positioning supported by familiar faces. Participants indicated that a combination of simple signage, minimal sound, dark colors, and elements of nature can create a calming location within a busy public transportation environment.
Stressful locations	In physical places, unwanted stimuli may include unpredictable people, undesirable lighting, and noise. Lighting types that made location unbearable, for example, fluorescent, or energy saving lighting. Lack of agency being able to cope with unexpected stimuli including workplaces, dentists and in the street. Environmental factors made locations uncomfortable, like temperature, salt water, and sandy beaches.

Table 9. Overview of Stressful Scenarios and Unpredictability

Sub-Code	Overview of <i>Stressful Scenarios</i> Results
Stressful scenarios	Autistic participants commented on unpredictable changes in plans that affected work choices and created stress. The stress of high-pressured jobs was also noted, including how a lack of routine and travel can create physical and cognitive problems. Respondents indicated that unpredictable nature of social communication tended to lead to the most stressful situations. Some respondents also noted the stress-related to confrontation. Interviewees, again, highlighted the stress associated with a variety of unwanted and unpredictable sounds. This can extend toward unwanted technological sounds, like notifications from mobile devices. Some noted the effects of stress when spending too long on their mobile devices. Participants commented on the stress associated with the imposition of rules regarding physical inactivity. Interviewees noted that scheduled assessments can be stressful. The replacement of humans for online services was suggested by interviewees as creating an increase in stress by reducing a routine approach to social interaction. Participants expressed discomfort being observed by others in working situations. Transport hubs were observed being overwhelming due to multisensory stimuli as well as interacting with others being stressed, while additional transport issues related to the unpredictability of their timetables.
Unpredictability	Interviewees commented on the unpredictability of social interaction and how it affects their choices in attempts to avoid other people. Participants commented on the unpredictable nature of people and how that affected personal choices, including public transportation options. Others noted that they plan as much of their day as they can, but things rarely go accordingly which can cause stress. Unexpected sounds and their associated stress were commented on by participants in environments out with their control. Respondents indicated that a lack of accuracy in information added to stress levels in environments that were already difficult, for example, hospital appointments, or flat sharing arrangements. Interviewees commented on unpredictable nature of meetings, where there was no prior indication of setup, and their attempts to reorganize things to make it more comfortable. Respondents noted perceived irrational changes in organization made shopping difficult for them. Other respondents commented on how other people ignoring rules causes them stress.

Table 10. Overview of Masking and Visibility

Sub-Code	Overview of Masking Results
Masking	Participants indicated the inability to identify personal problems before being diagnosed, for example, mental health issues. Others commented on problems caused by not being honest with their own thoughts and feelings. Respondents implied their natural behavior was not socially acceptable and made efforts to avoid it. Other participants commented on their discipline in learning and using masking behaviors, including social interaction in working environments. Interviewees commented on using distraction techniques to mask discomfort during social situations, for example, standing on one leg. Participants mentioned that they do not want to draw attention to themselves. Some noted the desire to fit in with what they perceived as normal. Respondents noted how disappointment in social experiences has shaped how they behave.
Visibility	Autistic participants commented on an awareness of being watched by others that can cause stress. Respondents implied that appearances should be maintained at the expense of true expression.

Table 11. Overview of Technology, Including Sub-Codes and Summarized Feedback, as Reported by Carers Concerning the People They Care for

Sub-Code	Overview of Technology Results
Technology	Hardware—compact cassettes, iPad. Software—favored regularly used apps.
Current use of technology	Hardware—tablet computers, iPads, MP3 players, headphones, smartphones, cassettes. Software—text to speech.
Current apps	Sensory apps—mainly for children, however, there is a desire for adults’ versions. Social stories—Pictorial. Other—YouTube, BBC. Special interest—Train Sim.
Potential for new technology	Need for familiarity—voices, stories, visual feedback, favored situations, familiar locations. Need for customization. Social stories, educational information, personalized visual feedback as positive distractors. Breathing, meditational exercises as means to focus. Hardware—headphones (for music and noise reduction), AR glasses, smartphones. Need for control and agency.
Reminder system	Need for hardware to recognize changes in the body, including heart rate and stress levels. Need for feedback system to highlight this—via audio feedback. Provide a step-by-step approach for guidance. Wearable and non-intrusive technology.

4.2.4 Locations. *Locations* encompass four sub-codes including *locations*, *familiarity*, *secure space*, and *stressful locations*. Participants discuss physical spaces visited or inhabited by autistic people in their care. They commented on familiarity being a positive major theme regarding physical spaces, digital spaces, and people, as well as unpredictable sensory stimuli being a stress trigger. The results are summarized in Table 14, with expanded results available in Supplementary Material C.

4.2.5 Stressful Scenarios. *Stressful Scenarios* encompass two sub-codes including *stressful scenarios* and *unpredictability*. Carers discussed situations that can occur that cause stress for autistic people in their care. The interviewees commented on stress caused by unpredictable sensory stimuli and unplanned inactivity, changes in routine, and problems in social situations due to emotional instability. The results are summarized in Table 15, with expanded results available in Supplementary Material C.

Table 12. Carers Overview of Coping Mechanisms, Emotional Regulation, Focus, Planning and the Agency, and Use of Music, Concerning the People They Care for

Sub-Code	Overview of <i>Coping Mechanisms</i> Results
Coping Mechanisms	Recreational and outdoor activities—for example, going to the zoo, day trips to the city, walking. Volunteer groups which help to facilitate activities. Music therapy, meditation, breathing exercises. Family as a main source of support.
Emotional regulation	Some autistic people are not good with verbal expression. Need for a familiar voice to reduce stress.
Focus	Using talking and storytelling as positive distractors. Watching videos, playing songs, and using technology can also be useful distractors. However, care needs to be taken to minimize obsessive use of technology and to encourage socialization.
Planning and agency	Plan and journal activities to reduce uncertainty and stress. Provide clear instructions to facilitate this. Detailing activities afterward can facilitate a greater sense of agency. Technology can aid in planning and agency, through scheduling apps on smartphones, and using headphones to control sound. Stress can be avoided at social events if planned with safe spaces identified beforehand. Personalized music choices aid in creating a sense of agency.
Use of music	Music and poetry from childhood can have a calming effect. Listening to and playing music can be beneficial for different occasions.

Table 13. Overview of Reactions, Lack of Body Awareness, Physical Reactions, Sensitivity, and Tactility, Regarding the People in Their Care

Sub-Code	Overview of <i>Reactions</i> Results
Reactions	External factors can affect emotional reactions, including unpredictable sounds and social interactions. Aggressive tendencies from some autistic people can be the result of unwanted or unplanned interactions. It can be difficult for autistic people to ask for appropriate help.
Lack of body awareness	Autistic people can find it difficult to recognize physiological and emotional changes and related stress levels. Autistic people can have a lack of concern for their own physical limitations and well-being.
Physical reactions	Repeated gestures and increasing levels of verbal expression can be indicators of stress. Other gestures include increased eye movement, rubbing of the eyes, and yawning. Fatigue can also be a sign of stress. Unexpected sounds and light can cause physical pain.
Sensitivity	Aural hypersensitivity, and the unpredictable nature of sounds, is commonly commented on as causing stress. Aural hyposensitivity can also be problematic, with autistic people desiring greater volume for watching TV or listening to music. Photosensitivity is also reported as a common problem in autistic people. Sunglasses and brimmed hats are commonly worn to address this.
Tactility	Autistic people can be hypersensitive regarding their skin, which can cause unexpected reactions, for example, experiencing pain from rainfall on their skin, or from some types of material. Objects designed to reduce overstimulation can cause problems themselves, for example, wearing glasses.

4.2.6 Masking. *Masking* encompasses two sub-codes including *masking* and *visibility*; however, no instances of *visibility* are reported here. Carers discussed an altered behavior in the people they care for, either due to social expectations or a desire to fit in. Participants commented on problems with acceptance of diagnosis, issues regarding expectations and trust, of others and of themselves. The results are summarized in Table 16, with expanded results available in Supplementary Material C.

Table 14. Overview of Locations, Familiarity, Secure Space, and Stressful Locations, Regarding the People in Their Care

Sub-Code	Overview of <i>Locations</i> Results
Locations	Respondents indicated that people in their care feel comfortable in familiar spaces and with familiar faces.
Familiarity	Routine locations, familiar people and voices, or habitual actions can create a sense of calm or relaxation for the autistic people in their care. Familiar activities including listening to music, or by following special interests, can create a sense calm.
Secure space	Entry and exit points need to be established immediately in new locations to feel secure. Others commented on the use of music, dim lighting, and avoiding crowds of people in creating a secure space for people in their care. Additionally, familiar digital spaces, like multiplayer games, can be a secure space for autistic people.
Stressful locations	Stress locations can be related to levels of uncertainty regarding personal sensitivity, for example, unwanted lights, smells, sound and touch, in doctors and dentists. Typically, unpredictable noise was reported as creating stressful locations, typically from people or animals, for example, in transport hubs, busy streets, or crowded areas.

Table 15. Overview of Stressful Scenarios and Unpredictability, Regarding the People in Their Care

Sub-Code	Overview of <i>Stressful Scenarios</i> Results
Stressful scenarios	Inactivity and unpredictability were associated with stressful scenarios, for example, waiting on buses or in airports. Additionally, a lack of routine can create stressful scenarios, for example, the closing of service centers that COVID lockdown created for service users. Similarly to stressful locations, any crowded area relating to the unpredictability of people can cause stress. Carers indicated that a lack of empathy for others' preferences can create problems where criticisms of others' actions would create tension.
Unpredictability	Interviewees commented on the unpredictable nature of sounds and its effect on the people they care for. Participants commented on the unpredictable emotional reactions of people in their care regarding social interactions, either external, or within the family home.

Table 16. Overview of Masking, Regarding the People in Their Care

Sub-Code	Overview of <i>Masking</i> Results
Masking	Participants noted that the people in their care did not accept their autism diagnosis and that this caused friction due to a lack of acceptance. Participants noted the inability of people in their care to apply their expectations of others to themselves. Participants comment on issues of trust in others' opinions. Respondents noted unrealistic expectations of the people in their care, and what they should earn and pay for.

5 Discussion

The results reiterate and further develop many of the themes derived from the first study [91]. This included how familiarity of people, locations, and routine aids in reinforcement of calming capabilities and reduction of stress in autistic individuals. Unpredictability, most commonly of sound, and the actions of people, causes stress and requires a more robust approach regarding the development of solutions. The use of positive distractors, whether audio, visual, or tactile, is currently used by participants in their own ways; however, there is much scope for development of technological solutions that could be used in *ad hoc* locations and scenarios as a form of self-intervention.

The analysis suggests several design implications derived directly from participants' lived experiences. Unpredictable auditory stimuli highlight the need for adaptive masking and ambient-audio

options that users can control. Familiarity emerged as a calming influence, indicating that personalized voices and color schemes may enhance comfort and trust. Fatigue linked to social unpredictability points to the value of planning tools that incorporate recovery time. Finally, the strong preference for control over sensory input underlines the importance of user-defined visual and haptic thresholds in any future system.

The results of this study indicated that there is interest in new self-intervention technologies to provide functionality to aid in stress management for autistic adults. The findings support other studies including SATORI, where the authors interviewed a small group consisting of four therapists and two autistic participants to design a mobile support tool for young autistic adults for use on smart devices [45]. The resultant design guidelines included: the use of breathing exercises to manage anxiety; real-time biometric monitoring to avoid overwhelming stimuli; the use of sensory stimuli as positive distractors; and developing a proposed app that can be used with existing devices. These guidelines reflect some of the findings in the present research; however, the authors noted that the study was small in scale and required implementation and testing for validation of its proposals. Hedges et al. [57] surveyed 472 autistic adolescents regarding technology use in managing stress. They reported widespread use of mobile technologies to support work, act as reminders and planning tools, or as ways to relax at home via audio, gaming, and video. The authors proposed that self-intervention technologies could be used to reduce anxiety and stress; however, they suggested there was a need to minimize potential obsessive use of devices. Contrary to this idea, other research indicated some obsessive behaviors displayed by autistic people tend to be related to preventing or reducing the discomfort that obsessive thought produces, suggesting this was a positive means of calming themselves [108]. Hoeberichts et al. [59] developed a mobile application aimed at supporting autistic adults where their app measures stress four times daily, generates an overview of the patients' stress, and suggests ways to cope, for example, "go for a walk," "do a breathing exercise" or "listen to some music." The study generated positive results; however, the authors contend that the results may have been influenced by the participants who helped to design the app. In addition, the app does not present real-time solutions but is based on post-analysis of voluntary participant information. Kalantari et al. [65] looked at wearable technology as a means for emotional regulation. The authors identified common stress triggers and reactions in a similar way to the present research, as well support types which included redirection, reminders, and planning, among the strategies. However, this article gathers information from parents of autistic people, rather than autistic participants, and looks to introduce a prototype smartwatch app with only visual recommendations for coping strategies at the time of publication.

The participatory emphasis aligns with Fletcher-Watson et al. [50], who argue that participatory methods can improve relevance, impact, and ethical standards in autism research. While Robertson and Simmons [122] offer firsthand accounts of sensory preferences and the importance of control over stimuli, which fits the discussion of sensory customization. Collectively, these comparisons underscore the novelty of integrating multimodal stress-management requirements into accessible AR frameworks for adults.

5.1 General Requirements

Data from this research suggest that the use of existing technology for new applications is favored over purchasing new devices, a theme reflected in the SATORI study [46]. Economic reasons were highlighted by autistic participants to support this idea, which have been further exacerbated in the past few years through rising living costs, COVID, and a subsequent rise in demand for cost-effective mental health services [86, 137]. Research has also indicated a resistance in autistic individuals to new experiences and technologies [103]; however, this was not reflected in participant responses. Related publications have identified ubiquitous smartphone use in autistic adults, as

well as other mobile technologies like tablet computers [70]; however, recent research has noted the need for more effective individualized applications in mobile and wearable technologies for autistic individuals (86) [73]. It is proposed that a mobile application be developed that allows use of target users' existing hardware, including headphones for auditory stimuli. Sensors in existing smartphones, including older models, could form a cost-effective basis of detecting and analyzing biometric data of the target group. Previous autism-related studies have made use of pedometers [76], accelerometers [78], and gyroscopes [71] to gather data. These sensors could be used to pick up different forms of stimming, similar to a study by Sarker et al. [127] where a smartwatch-based system was used to detect gestural movements of adults who mimicked the movements of autistic children. Making use of existing hardware could address economic issues regarding the purchasing of new hardware, as well as resources required to learn a completely new system. Available studies on the efficacy of mobile applications for autistic people show promise, however, additional well-designed, and larger scale studies, are recommended [98].

Familiarity was one of the themes identified as a coping mechanism in the first phase of the study. Development of an app using existing hardware could utilize familiarity, regarding the routine use of technology, or through interactions with familiar people and processes. Related research using caregivers' voices as notifications via a smartwatch to autistic children, suggested that using the voice of known instructors or caregivers could lead to better performance when using apps [159, 160]. This could be achieved using contemporary voice banking technologies such as Apple's Personal Voice, where synthesized voices, based on your own, or your family and friends, can be used within assistive communication apps or in-person conversations on mobile devices [6].

Respondents in the present study noted that an application would require an accessible setup and interface, offering simple instructions to support its use for the target group's subjective needs. Customizability was indicated as key to any new technology being adopted and regularly used, which has been supported by previous research [37], including potential for the use of AI in personalizing content for individual use. Rehman et al. [121] proposed recommendations in future development of mobile apps for autistic people including progress tracking, personalized content delivery, automated reasoning, image recognition, and Natural Language Processing. The results suggest that user profiling and real-time data updates, based on users' chosen biometrics, could be beneficial in addressing their specific needs. In addition, autistic participants suggested that a planning tool could aid in daily organization and provide reminders, as well as build in recovery time for potentially unavoidable stressful situations. A contemporary app designed by neurodiverse people, named Thruday [147], provides many useful features that integrates planning, routines, mood management, uses AI, and allows the user to invite friends or family to assist could be used a useful guide for any new self-intervention.

5.2 Auditory Sensitivity

Sound has frequently been referred to as the most common sensory stressor for autistic people [81, 140]. Reflecting previous research regarding stress-related unpredictability in autism [54, 90], carers and autistic participants identified unpredictable scenarios involving people that created stress. Previous research has identified the difficulties for autistic people in auditory processing within noisy environments [16]. Some autistic people employ noise-canceling technologies (passive/active) or make use of masking techniques, including white noise, rain or other sound effects, which are available through a variety of apps [162]. Alternatively, ambient music could be used as a means of masking out unwanted background noise, or as a means to aid in reduction of stress levels. Previous research into the use of consonant ambient music, as well as tempo-specific rhythm interventions, has demonstrated a positive reduction on the anxiety levels of autistic children [20]. Subsequently, multiple techniques to explore audio sensitivity must be a priority within any new system. Sound

masking techniques are well-established within the workplace in efforts to address persistent tension between public social interaction and private undisturbed work in many open-plan offices [27, 60]. A 2016 study supported the argument that an attentional deficit, rather than a perceptual deficit, affected the ability of autistic individuals to discriminate speech from background noise [43]. Results suggest that real-time external monitoring of frequencies and dynamics could be facilitated by using current smartphone technologies, which could then automatically be addressed using masking techniques that would create a suitable ambient backdrop of the users choosing.

Other forms of effective audio attenuation for autistic people have been explored, including the use of ears muffs, and via noise-canceling functionality [62, 113]. Noise cancelation could also be included within an app and employed dependent on the available components within a user's smartphone. An optimal approach would allow continual monitoring of external audio using reinforcement learning as a viable approach. Alternatively, machine learning could also be employed when reinforcement learning does not provide a suitable solution, as previously demonstrated via positive results in a chatbot deep learning system, designed for autistic children [79]. Recent work has investigated physiological, behavioral, and neural responses to reinforcers to uncover differences in motivation and learning in autism, which can be synthesized and used as early intervention for autistic people [128]. Reinforcement learning, a method used within AI, has been employed in a recent prototype app to assist autistic students' educational experiences with positive, but limited results [145]. The mobile app was created with the aim of assisting school aged educators in tracking students' progress, as well as providing motivators for the students themselves when needed, to complete a task. However, the suggested motivators were not customized for each student, a customer space had to be built for observation, and suggestions were based on a generalized algorithm. Nonetheless, this study does demonstrate the potential for reinforcement learning as a basis for a new app. Any General Data Protection Regulation [114] concerns regarding the security of personal data gathered could be alleviated by using local neural processing units within existing mobile devices [64], thereby removing the need for transference of data beyond the device itself.

Regarding audio management, unwanted user-defined frequency ranges could be predetermined during a calibration phase of a new application, updated continually as identified. A listen through mode would allow users to be aware of the environment but block out unwanted frequencies and sounds that exceed a desired volume threshold. Listen through (or hear through) modes are currently used within a variety of headphone manufacturers to allow external sound to be heard through the headphones, when desired, as you listen to your music [104]. The technology uses a built-in microphone to enhance external sound and focus on people's voices when desired [10]. In addition, a context-aware system could also feed into the application where, for example, real-time detection of spectral and dynamic content, or Wi-Fi signals, could determine the location of the device and associated sounds, and switch to the appropriate mode, like workplace, university, or home. Research has demonstrated the ability to utilize context-aware systems using active audio sensing, where a mobile device speaker emits an ultrasonic signal, and the microphone detects its reflected signal from ambient objects that can help determine spatial size and surrounding materials [77]. Tang et al. developed a cheap context-aware mobile app for autistic adults to aid in daily activities like cooking and cleaning [144]. The authors utilized location-based infrared and Bluetooth wireless sensors that were detected by a smartphone app, which provided multimodal feedback available for the user. For example, video and audio explanations were provided including a "Tell me what to do" button that could give users an audio instruction. The study presented positive results but was limited to a location-based scenario, with the authors suggesting body-based sensors to detect and use biometric data for future development. Alternatively, visually aware systems could

be interlinked with a mobile app in the same way that museums detect visitor movement to trigger appropriate related audio guides [129].

5.3 Multimodal Positive Distractors

The results have demonstrated that the use of personal distractors in uncomfortable situations can be beneficial for autistic adults. The data indicate that a digital version of this could be advantageous for the target group. This could be presented in a multimodal way, via audio, visual, or haptic feedback, determined by the contextual nature of a given scenario, and preferences of the user. Via AI reinforcement learning, a new system could record personal training data as a baseline, while consistently monitoring gestural movement, to be used to provide positive solutions to address any stressful situations identified. Gesture-controlled apps are currently available for Android and iOS systems aimed at users with accessibility needs and could be adapted within any smart device to recognize positive or negative gestures and react accordingly for autistic users [3, 97]. Links to user favorites, social media choices, location tracking (including Wi-Fi zones), time of day, or any system that provides positive and negative qualitative experiences could be tracked to build a user profile of choices and reactions. Microphones within mobile devices can monitor and record audio experiences that add to the profile detail to build scenario-based modes [158]. External changes in sound pressure levels and frequencies could look at related choices and use the users previously recorded reactionary data to pre-empt their requirements. For example, audio-based distractors could employ preferential music that relates to calmer time periods or that signifies calming locations or familiar people.

A study conducted by Parmar et al. [107] looked at visual sensory experiences from the perspective of autistic adults, found that moderating factors, such as reduced sensory inputs, reduced sensory intensity, and predictable environments helped in reducing stress. Additionally, lighting, in terms of brightness and color temperature, was identified as an important factor in reduction of stress via the use of colored overlays or lenses, supporting previous research into coping mechanisms of overwhelming sensory experiences of autistic adults [123, 136]. Results of the present study reflect some of this and have indicated that participants customization of visually based distractors could be based upon simple color choices, still images, animations, or more pattern-based distractors, as coping mechanisms in different scenarios, within a new app. Alternatively, restriction of visual stimuli could also help to block out overwhelming stimuli; however, this would require a head-mounted display to be effective, which are currently less commonly used than smart devices. Other ways that visually augmentation could be beneficial include educational ways to provide prompts, visualize objects or locations, and provide detailed steps involved in unknown processes (see Section 5.5).

Previous research has seen the design of haptic interfaces and tangible virtual environments for autistic children to simulate environments, learn, as well as to encourage interaction and play [32, 110]. Interviewees suggested that haptic feedback could be an effective and subtle way to trigger notifications concerning their health but also to provide schedule reminders. They suggested arm bands or currently used smartwatches as a means to do so. A related study by Kelling et al., developed a haptic aid aimed at reducing stress for neurotypical participants via a sleeve embedded with actuators, a heart rate monitor, and a smartwatch [68], demonstrated a decrease in heart rate via haptic stimulation within its participants. A similar approach could be used for autistic participants.

Many mobile application hardware devices are utilized for fitness tracking as well as tracking oxygen levels [51]. Studies researching the effectiveness of wearable health tracking devices have highlighted issues regarding a lack of appropriate updates for devices, and inaccuracies in data, related to devices' inability to monitor biometric data through darker than white skin tones [119, 156]. However, the authors also discuss developments in motion artifact reduction using

multi-wavelength approaches, and the effects of skin temperature on optical pulse detection “photoplethysmography” sensing, as well as the need for additional studies that investigate the combined effects of these factors.

5.4 Biometric Monitoring

Participants commented on issues regarding sensory overload through hypersensitivity to external stimuli. Additionally, issues were identified regarding hypo sensitivity, which has led to issues with physical ill health, and long-term mental health problems. This relationship which has been previously identified in autism research where negative emotionality, depression, and anxiety symptoms were identified in autistic children and adolescents [17, 112]. Subsequently, it is proposed that an app could monitor physical biometrics in real-time, for example, heartrate, eye movement, stimming, user-defined gestures, hydration levels, or changes in voice. The users would be alerted through audio or haptic feedback as an indication that attention is required. Security issues of any potential AR application need to be considered, especially when dealing with personal information. A recent review highlighted some of the issues that exist with current cyber security protocols which rely on traditional cryptographic protocols such as Transport Layer Security and Virtual Private Networks [13]. None of these secure communication solutions ensures the preservation of the integrity of the captured data end-to-end, as such protocols are known to have Man-in-the-Middle vulnerabilities. The authors suggested that further research and development of security-based tools for AR should be considered. Subsequently, development of new AR-based prototypes should ensure that data is kept within the system until such times as a more robust solution for securing data exists.

5.5 AR

Despite some participant feedback regarding economic reasons for not investing in new technology, the interview data demonstrated an interest in the potential for using multimodal digital stimuli via augmented or mixed reality. This also reflects the findings of the first phase of the study, where caregivers and autistic participants expressed an interest in using AR technology [91]. Interactive AR applications designed for autistic users have demonstrated efficacy through assisting children in teaching and learning, including learning vocabulary [69, 125] and through simulation of hygiene related activities, or being in public spaces like supermarkets (135) [157]. A 2020 review highlighted that social communication skills were the most targeted within AR applications and that computer, smartphone, and smart glasses were the most frequently used technologies [70]. The results of the studies demonstrated that AR benefitted autistic children in learning skills; however, due to a lack of available studies, the authors suggested that future research address the effectiveness of AR among greater numbers of participants, as well as exploring different technologies supporting AR for intervention purposes. For example, the authors commented on the lack of projection-based technologies used, due to their expense, and suggested future studies investigate the use of head-mounted AR hardware like Microsoft HoloLens [96], Magic Leap [85], or Oculus Rift, now named the Meta Quest [93].

Future work will involve participatory prototyping of these requirements and validation of stress-reduction efficacy in real-world contexts. These steps will extend the current findings into deployable assistive technologies. In terms of how AR could be used for self-intervention for participants in the present study, it is proposed that augmented audio could be used to support day-to-day activities in real time, for example, providing shopping or public transport information. Sound effects from nature could be employed to counter artificial and sensory overwhelming locations, auditioning possible solutions to minimize the impact of sensory overload, based on the user’s reinforced learning system. Alternatively, sounds that belong in a given environment could

be used to mask undesirable sounds, where they could be minimized within overwhelming locations like busy streets, or in the workplace. Such systems already utilize situational awareness capabilities, vital for performance and error prevention in safety critical domains [155], for example, so that music being played by a user can be bypassed in favor of the sound of emergency services within immediate physical proximity [92]. A recent study in Context-Aware Sound Effect Generation in AR utilizes computer vision models and a large language model to generate text descriptions that incorporate context information of user, virtual object and real-world environment data [142]. Subsequent retrieval of appropriate sound effects were played back with AR visuals for a contextual experience. Mariette [87] discusses the advantages of using spatial audio in AR applications, highlighting efficiencies in understanding of semantic and directional data. Using a similar system, sound could also be used to guide users under duress to somewhere safe using a Tinkerbelle Effect [44] where sounds could be used via rhythmic and spectral processing to produce a directional auditory illusion, leading them to a safer space.

Familiar voices could also be employed as a digital coping mechanism, as a way to introduce calm. This could be extended to tell either recorded stories or use AI technology to apply familiar voices to other narratives, or as a chatbot. Virtual voice assistants have demonstrated positive effects on the learning capabilities of autistic children [126], while open sourced coding allows the customization of virtual voice assistants based on recorded audio files from any source [120].

In addition to monitoring biometric data, an AR app could monitor stimming and provide gesture control functionality to entrain with multimodal stimuli, prompting the user in a focused and guided manner as a positive distractor, with the aim of reducing heart rate and associated stress levels. Similarly, visual support could be used via text, image, video, or animations, to support identification of objects or processes, provide navigation, project familiar faces or preferred stimuli onto AR glasses. A 2020 review of the use of AR in autism-based intervention studies identified positive effects on improving different domains such as social interaction, social communication skills, verbal and non-verbal communication, facial emotion recognition procedures, attention skills, or functional life in children and adolescents with autism [19]. Regarding the technology used in the different interventions, three studies were carried out with a smartphone, six studies used smart glasses, and 11 studies used multiple devices (AR device with screens and helmet, computer games, or computer tablets). Currently, AR glasses are in use and becoming more popular with new developments from the major companies including Apple [7] and Meta [94]. Ray Bans [82] are developing new technology that will soon be available for commercial use that would present a more acceptable form of AR that could draw less attention to its wearer in public, supporting the present results regarding *Masking*. AR glasses could also allow customized personal biometric data to be projected for user feedback or to reduce unwanted light levels for photosensitive users.

5.6 Additional Contributions

This study presents data from both autistic participants and caregivers within the same paper, supporting publications that adopt a multiperspective design. While contemporary research on autistic adults typically gathers data from either autistic participants [33, 136] or caregivers [118] [65], this study integrates both perspectives to provide a more comprehensive understanding. Mattys et al. [89] interviewed young autistic adults and mental health professionals, identifying key themes such as searching for balance and negotiating autism, as well as seeking suitable surroundings in different areas of life. While Espinosa and Escobedo conducted a small-scale study involving therapists and autistic adults in their research on stress, they only interviewed four therapists and two adults [46].

This study provides a qualitative review of technologies used by autistic participants in past and present contexts, exploring their role in managing stress and daily challenges. It examines why

certain technologies were ultimately discarded and identifies desired features for new developments. Previous reviews of technologies used by autistic individuals have largely been meta-reviews [5, 26, 53, 139, 143, 149] or surveys [65, 115], which may not provide the same level of detailed insight needed for designing effective new technologies.

Building on these findings, this research proposes a design for an app that incorporates audio and visual distractors, informed by participant feedback on effective coping mechanisms. Participants identified music, nature sounds, and controlled visual stimuli as useful strategies for emotional regulation (see Sections 4.1.2 and 4.1.4). By structuring these elements into an interactive and adaptable digital format, the proposed app aims to provide a flexible and accessible tool for managing sensory and emotional challenges.

The app's design includes customizable auditory elements, such as white noise, nature sounds, and harmonic tones, alongside visual elements like gentle animations, color gradients, and interactive patterns (see Sections 5.2 and 5.3). Since sensory predictability was noted as a key factor in stress management, these features will allow users to adjust sound frequencies, visual speed, and intensity to match their preferences. Additionally, the app could include real-time environmental adjustments, such as dimming the screen in low-light settings or modifying audio output based on background noise. Future iterations could integrate biometric-based feedback, adapting stimuli based on real-time physiological data such as heart rate variability or skin conductance (see Section 5.4).

Usability considerations are critical, particularly regarding accessibility, personalization, and sensory sensitivities. The app would feature color customization, volume modulation, and a gradual fade-in/fade-out function to prevent sudden sensory changes. Users could save personalized profiles, allowing quick access to preferred settings based on different stress levels or environments (see Section 5.3). Integration with assistive technologies and compatibility with external devices, such as noise-canceling headphones or screen readers, would further enhance usability.

By consolidating participant insights, this study outlines preliminary requirements that could serve as a foundation for researchers and developers working on alternative stress-management apps for autistic adults. Prioritizing customization, sensory regulation, and accessibility, this proposal contributes to the development of technologies that effectively support autistic users in managing their daily lives.

6 Conclusions

The current work presents a novel dataset of qualitative data from five caregivers and 15 autistic individuals. The dataset extends the findings from an initial survey highlighting some common themes, including familiarity, unpredictability, the use of positive distractors as coping mechanisms, and an interest in AR technologies. Auditory sensitivity remains the primary source of stress-related reactions.

Regarding the overarching goal of mitigating the potentially debilitating effects of stress, proposals have been made regarding future work and the potential for prototype developments for use on users' existing mobile devices and headphones, allowing the use of audio noise canceling or listen-through capabilities when required. Monitoring of real-time biometric data that can measure and pre-empt the impact of stress is proposed by presenting the user with appropriate distractors, or by providing digital assistance dependent on the given situation, based upon learned behaviors via an AI reinforcement learning method. User feedback should be customized using audio, visual, and haptic feedback. Audio could be used to mask unwanted sounds within stressful locations or augment existing soundscapes with more pleasing sounds determined by the user. The use of familiar voices could also provide a calming distraction. Similarly, visual stimuli could be presented as a positive distractor or to augment existing scenarios with digital overlays via mobile device

screens, or via head-mounted smart glasses. Alternatively, head-mounted AR systems would also allow for the restriction of overwhelming visual stimuli, as well as providing color overlays or reducing light sensitivity in photosensitive users. Haptic feedback is proposed as a subtle biometric reminder system and could be used in conjunction with audio-visual feedback stimulus to aid in stimulating or gestural entrainment methods to reduce stress.

Future work aims to advance the proposals in this study to design, develop, and test separate audio, visual, and haptic prototypes, as well as combined versions of these, toward a consolidated and accessible functionality, that allows effective customization for autistic users.

The complete anonymized interview dataset can be obtained by contacting the authors.

Acknowledgments

The Authors would like to acknowledge the School of Computing, Engineering, and the Built Environment for their financial support, to Autistica and Differabled Scotland for their assistance in the recruitment of the study, and to everyone who took part in the study.

References

- [1] Yurena Alonso-Esteban and Francisco Alcantud-Marín. 2021. Autism spectrum disorders and anxiety: measurement and treatment. In *Neural Engineering Techniques for Autism Spectrum Disorder*. Elsevier, 315–332. DOI: <https://doi.org/10.1016/B978-0-12-822822-7.00015-6>
- [2] Amazing Things Happen. 2024. Amazing Things Happen! - YouTube. Retrieved May 3, 2024 from <https://www.youtube.com/watch?v=Ezv85LMFx2E>
- [3] American Foundation for the Blind. 2024. Operating Your iPhone or iPad with Gestures | American Foundation for the Blind. Retrieved June 24, 2024 from <https://www.afb.org/blindness-and-low-vision/using-technology/cell-phones-tablets-mobile/apple-ios-iphone-and-ipad-1>
- [4] American Psychiatric Association. 2013. *Diagnostic and Statistical Manual of Mental Disorders* (5th. ed.). American Psychiatric Publishing, Arlington, VA. DOI: <https://doi.org/10.1016/B978-0-12-809324-5.05530-9>
- [5] Angeliki Sideraki. 2024. Exploring the therapeutic potential of virtual reality and augmented reality in social skills training for individuals with autism spectrum disorder. *World Journal of Advanced Research and Reviews* 23, 3 (September 2024), 2995–3004. DOI: <https://doi.org/10.30574/wjarr.2024.23.3.3002>
- [6] Apple Inc. 2024. Create a Personal Voice on Your iPhone, iPad or Mac – Apple Support (UK). Retrieved June 27, 2024 from <https://support.apple.com/en-gb/104993>
- [7] Apple Inc. 2024. Augmented Reality - Apple (UK). Retrieved June 26, 2024 from <https://www.apple.com/uk/augmented-reality/>
- [8] Zinat Ara, Amrita Ganguly, Donna Peppard, Dongjun Chung, Slobodan Vucetic, Vivian Genaro Motti, and Sungsoo Ray Hong. 2024. Collaborative job seeking for people with autism: Challenges and design opportunities. In *Proceedings of the Conference on Human Factors in Computing Systems*, ACM. DOI: <https://doi.org/10.1145/3613904.3642197>
- [9] ASAN. 2024. Identity-First Language. Autistic Self Advocacy Network. Retrieved January 11, 2024 from <https://autisticadvocacy.org/about-asan/identity-first-language/>
- [10] Audio-Technica. 2024. Understanding the Hear-Through Feature on Audio-Technica Headphones | Audio-Technica. Retrieved June 24, 2024 from <https://www.audio-technica.com/en-eu/support/understanding-the-hear-through-feature-on-audio-technica-headphones>
- [11] Autistica. 2022. Home - Autism | Autistica. Autistica. Retrieved April 6, 2022 from <https://www.autistica.org.uk/>
- [12] Michael B. Bakan. 2015. The musicality of stimming: Promoting neurodiversity in the ethnomusicology of autism. *MUSICultures* 41, 2 SE-Articles (May 2015). Retrieved from <https://journals.lib.unb.ca/index.php/MC/article/view/22914>
- [13] Tianyou Bao, Hurriyet Ok, T. Bao, and H. Ok. 2021. Secure augmented reality (AR) for telehealth and emergency medical services (EMS): A survey. Cyber Security and Privacy Research Institute, The George Washington University.
- [14] M. Grace Baron, June Groden, Gerald Groden, and Lewis P. Lipsitt. 2016. *Stress and Coping in Autism*. Oxford University Press. DOI: <https://doi.org/10.1093/med-psych/9780195182262.001.0001>
- [15] BBC. 2024. BBC Two - Inside Our Autistic Minds. Retrieved May 3, 2024 from <https://www.bbc.co.uk/programmes/p0bbnh47>
- [16] Rachel Bellamy, Howard Ring, Peter Watson, Andrew Kemp, Giles Munn, and Isabel C. H. Clare. 2021. The effect of ambient sounds on decision-making and heart rate variability in autism. *Autism* 25, 8 (November 2021), 2209–2222. DOI: <https://doi.org/10.1177/13623613211014993>

- [17] A. Ben-Sasson, S. A. Cermak, G. I. Orsmond, H. Tager-Flusberg, M. B. Kadlec, and A. S. Carter. 2008. Sensory clusters of toddlers with autism spectrum disorders: Differences in affective symptoms. *The Journal of Child Psychology and Psychiatry* 49, 8 (August 2008), 817–825. DOI : <https://doi.org/10.1111/j.1469-7610.2008.01899.x>
- [18] Esma Mansouri Benssassi, Juan Carlos Gomez, Louanne E. Boyd, Gillian R. Hayes, and Juan Ye. 2018. Wearable assistive technologies for autism: Opportunities and challenges. *IEEE Pervasive Computing* 17, 2 (2018), 11–21. DOI : <https://doi.org/10.1109/MPRV.2018.022511239>
- [19] Carmen Berenguer, Inmaculada Baixauli, Soledad Gómez, María de El Puig Andrés, and Simona De Stasio. 2020. Exploring the impact of augmented reality in children and adolescents with autism spectrum disorder: A systematic review. *International Journal of Environmental Research and Public Health* 17 (2020), 1–15. DOI : <https://doi.org/10.3390/ijerph17176143>
- [20] Dorita S. Berger. 2012. Pilot study investigating the efficacy of tempo-specific rhythm interventions in music-based treatment addressing hyper-arousal, anxiety, system pacing, and redirection of fight-or-flight fear behaviors in children with autism spectrum disorder (ASD). *Journal of Biomusical Engineering* 2 (2012), 1–15. DOI : <https://doi.org/10.4303/jbe/m110902>
- [21] Geetha Bharathi, Kaavya Jayaramayya, Venkatesh Balasubramanian, and Balachandar Vellingiri. 2019. The potential role of rhythmic entrainment and music therapy intervention for individuals with autism spectrum disorders. *Journal of Exercise Rehabilitation* 15, 2 (2019), 180–186. DOI : <https://doi.org/10.12965/jer.1836578.289>
- [22] Ananda Balayogi Bhavanani. 2018. Addressing autism spectrum disorder through yoga as a complementary therapy. *SBV Journal of Basic, Clinical and Applied Health Science* 1, A3 (June 2018), 3–7. DOI : <https://doi.org/10.5005/jp-journals-10082-01123>
- [23] Lauren Bishop-Fitzpatrick, Carla A. Mazefsky, Nancy J. Minshew, and Shaun M. Eack. 2015. The relationship between stress and social functioning in adults with autism spectrum disorder and without intellectual disability. *Autism Research* 8, 2 (April 2015), 164–173. DOI : <https://doi.org/10.1002/aur.1433>
- [24] Melissa H. Black, Rebecca Kuzminski, Jing Wang, Jessica Ang, Cheryl Lee, Siti Hafidzuddin, and Sarah McGarry. 2024. Experiences of friendships for individuals on the autism spectrum: A scoping review. *Review Journal of Autism and Developmental Disorders* 11 (2024), 184–209. DOI : <https://doi.org/10.1007/s40489-022-00332-8>
- [25] K. Boshoff, D. Gibbs, R. L. Phillips, L. Wiles, and L. Porter. 2016. Parents’ voices: ‘why and how we advocate’. A meta-synthesis of parents’ experiences of advocating for their child with autism spectrum disorder. *Child Care, Health and Development* 42, 6 (November 2016), 784–797. DOI : <https://doi.org/10.1111/cch.12383>
- [26] Lal Bozgeyikli, Andrew Raij, Srinivas Katkoori, and Redwan Alqasemi. 2018. A survey on virtual reality for individuals with autism spectrum disorder: Design considerations. *IEEE Transactions on Learning Technologies* 11, 2 (April 2018), 133–151. DOI : <https://doi.org/10.1109/TLT.2017.2739747>
- [27] Joeri Bruyninckx. 2023. Tuning the office sound masking and the architectonics of office work. *Sound Studies* 9, 1 (2023), 64–84. DOI : <https://doi.org/10.1080/20551940.2022.2162765>
- [28] Richard A. Bryant, Allison G. Harvey, Suzanne T. Dang, Tanya Sackville, and Chris Basten. 1998. Treatment of acute stress disorder: A comparison of cognitive-behavioral therapy and supportive counseling. *Journal of Consulting and Clinical Psychology* 66, 5 (October 1998), 862–866. DOI : <https://doi.org/10.1037/0022-006X.66.5.862>
- [29] Stéphanie Carlier, Sara Van der Paelt, Femke Ongenaes, Femke De Backere, and Filip De Turck. 2020. Empowering children with ASD and their parents: Design of a serious game for anxiety and stress reduction†. *Sensors* 20, 4 (2020). DOI : <https://doi.org/10.3390/s20040966>
- [30] Ferrán Catalá-López, Manuel Ridao, Isabel Hurtado, Amparo Núñez-Beltrán, Ricard Gènova-Maleras, Adolfo Alonso-Arroyo, Aurelio Tobías, Rafael Aleixandre-Benavent, Miguel A. Catalá, and Rafael Tabarés-Seisdedos. 2019. Prevalence and comorbidity of autism spectrum disorder in Spain: study protocol for a systematic review and meta-analysis of observational studies. *Systematic Reviews* 8, 1 (December 2019), 141. DOI : <https://doi.org/10.1186/s13643-019-1061-1>
- [31] Francisco Cavas-Martínez, Manuel D. Marín Granados, Ramón Mirálbes Buil, and Oscar D. de-Cózar-Macías (Eds.). 2023. *Advances in Design Engineering III*. Springer International Publishing, Cham. DOI : <https://doi.org/10.1007/978-3-031-20325-1>
- [32] Phoebe Chen, Alfredo Cuzzocrea, Xiaoyong Du, Orhun Kara, Ting Liu, Krishna M. Sivalingam, Dominik Słezak, Takashi Washio, Xiaokang Yang, Simone Diniz, et al. 2019. Editorial Board Members. Retrieved from <http://www.springer.com/series/7899>
- [33] Andrew M. Colombo-Dougovito, A. Josephine Blagrove, and Sean Healy. 2021. A grounded theory of adoption and maintenance of physical activity among autistic adults. *Autism* 25, 3 (April 2021), 627–641. DOI : <https://doi.org/10.1177/1362361320932444>
- [34] Julia Cook, Laura Crane, Laura Hull, Laura Bourne, and William Mandy. 2022. Self-reported camouflaging behaviours used by autistic adults during everyday social interactions. *Autism* 26, 2 (February 2022), 406–421. DOI : <https://doi.org/10.1177/13623613211026754>

- [35] John W. Creswell and Cheryl N. Poth. 2016. *Qualitative Inquiry and Research Method: Choosing among Five Approaches* (4th. ed.). Sage Publications, Inc.
- [36] Elizabeth K. Cridland, Sandra C. Jones, Peter Caputi, and Christopher A. Magee. 2015. Qualitative research with families living with autism spectrum disorder: Recommendations for conducting semistructured interviews. *Journal of Intellectual & Developmental Disability* 40, 1 (January 2015), 78–91. DOI: <https://doi.org/10.3109/13668250.2014.964191>
- [37] Antonina Dattolo and Flaminia L. Luccio. 2017. Accessible and usable websites and mobile applications for people with autism spectrum disorders: A comparative study. *ICST Transactions on Ambient Systems* 4, 13 (May 2017), 152549. DOI: <https://doi.org/10.4108/eai.17-5-2017.152549>
- [38] Nyaz Didehbani, Tandra Allen, Michelle Kandalaft, Daniel Krawczyk, and Sandra Chapman. 2016. Virtual reality social cognition training for children with high functioning autism. *Computers in Human Behavior* 62 (2016), 703–711. DOI: <https://doi.org/10.1016/j.chb.2016.04.033>
- [39] Differabled. 2019. DIFFERabled Scotland. Retrieved February 15, 2024 from <https://www.differabledscotland.co.uk/about>
- [40] Renee R. Dijkhuis, Tim Ziermans, Sophie van Rijn, Wouter Staal, and Hanna Swaab. 2019. Emotional arousal during social stress in young adults with autism: Insights from heart rate, heart rate variability and self-report. *Journal of Autism and Developmental Disorders* 49 (2019), 2524–2535. DOI: <https://doi.org/10.1007/s10803-019-04000-5>
- [41] Melissa Sandra Duchene. 2015. Maternal experiences of parenting children with autism spectrum disorder: A qualitative analysis. *ProQuest Dissertations and Theses* 44, 5 (2015), 314. Retrieved from http://proxy.bc.edu/login?url=http://search.proquest.com/docview/1707355040?accountid=9673%0Ahttp://bc-primo.hosted.exlibrisgroup.com/openurl/BCL/services_page?url_ver=Z39.88-2004&rft_val_fmt=info:ofi/fmt:kev:mtx:dissertation&genre=dissertations+%26+thes
- [42] Katerina M. Dudley, Mark R. Klinger, Allison Meyer, Patrick Powell, and Laura G. Klinger. 2019. Understanding service usage and needs for adults with ASD: The importance of living situation. *Journal of Autism and Developmental Disorders* 49, 2 (February 2019), 556–568. DOI: <https://doi.org/10.1007/s10803-018-3729-0>
- [43] William A. Dunlop, Peter G. Enticott, and Ramesh Rajan. 2016. Speech discrimination difficulties in high-functioning autism spectrum disorder are likely independent of auditory hypersensitivity. *Frontiers in Human Neuroscience* 10 (August 2016), 12. DOI: <https://doi.org/10.3389/fnhum.2016.00401>
- [44] Frank H. Durgin. 2002. The Tinkerbell effect motion perception and illusion. *Journal of Consciousness Studies* 9, 5–6 (2002), 88–101.
- [45] Marcela A. Espinosa and Lizbeth Escobedo. 2023. Designing a stress and anxiety support tool to help young adults with autism in daily living. *Interacting with Computers* 35, 2 (August 2023), 83–90. DOI: <https://doi.org/10.1093/iwc/iwad012>
- [46] Marcela Espinosa and Lizbeth Escobedo. 2021. SATORI: A support tool for young adults with autism to self-manage stress and anxiety. In *X Latin American Conference on Human Computer Interaction*, ACM, New York, NY, 1–4. DOI: <https://doi.org/10.1145/3488392.3488394>
- [47] Rainart Fayette and Caroline Bond. 2018. A systematic literature review of qualitative research methods for eliciting the views of young people with ASD about their educational experiences. *European Journal of Special Needs Education* 33, 3 (May 2018), 349–365. DOI: <https://doi.org/10.1080/08856257.2017.1314111>
- [48] Erin Felephchuk. 2021. Stimming, improvisation, and COVID-19: (Re)negotiating autistic sensory regulation during a pandemic. *Disability Studies Quarterly* 41, 3 (September 2021). DOI: <https://doi.org/10.18061/DSQ.V41I3.8426>
- [49] Michael B. First. 2013. *DSM-5® Handbook of Differential Diagnosis*. DOI: <https://doi.org/10.1177/APPI BOOKS.9781585629992>
- [50] Sue Fletcher-Watson, Jon Adams, Kabie Brook, Tony Charman, Laura Crane, James Cusack, Susan Leekam, Damian Milton, Jeremy R. Parr, and Elizabeth Pellicano. 2019. Making the future together: Shaping autism research through meaningful participation. *Autism* 23, 4 (May 2019), 943–953. DOI: <https://doi.org/10.1177/1362361318786721>
- [51] Future US Inc. 2024. Best Smartwatches that Can Measure Blood Oxygen Saturation Levels | Android Central. Retrieved June 25, 2024 from <https://www.androidcentral.com/best-smartwatches-can-measure-blood-oxygen-saturation-levels>
- [52] D. A. Garcia-Villamizar and J. Dattilo. 2010. Effects of a leisure programme on quality of life and stress of individuals with ASD. *Journal of Intellectual Disability Research* 54, 7 (July 2010), 611–619. DOI: <https://doi.org/10.1111/j.1365-2788.2010.01289.x>
- [53] Tina R. Goldsmith and Linda A. Leblanc. 2004. Use of technology in interventions for children with autism. *Journal of Early and Intensive Behavior Intervention* 1 (2004), 166–178.
- [54] Marie Gomot and Bruno Wicker. 2012. A challenging, unpredictable world for people with autism spectrum disorder. *International Journal of Psychophysiology* 83 (2012), 240–247. DOI: <https://doi.org/10.1016/j.ijpsycho.2011.09.017>
- [55] Temple Grandin. 2009. Visual abilities and sensory differences in a person with autism. *Biological Psychiatry* 65, 1 (2009), 15–16. DOI: <https://doi.org/10.1016/j.biopsych.2008.11.005>
- [56] Greg Guest, Arwen Bunce, and Laura Johnson. 2006. How many interviews are enough? An experiment with data saturation and variability. *Field Methods* 18, 1 (2006), 59–82. DOI: <https://doi.org/10.1177/1525822X05279903>

- [57] Susan H. Hedges, Samuel L. Odom, Kara Hume, and Ann Sam. 2018. Technology use as a support tool by secondary students with autism. *Autism* 22, 1 (January 2018), 70–79. DOI: <https://doi.org/10.1177/1362361317717976>
- [58] A. Hillier, J. Kopeck, N. Poto, M. Tivarus, and D. Q. Beversdorf. 2015. Increased physiological responsiveness to preferred music among young adults with autism spectrum disorders. *Psychology of Music* (2015), 0305735615576264-. DOI: <https://doi.org/10.1177/0305735615576264>
- [59] Kirsten Hoeberichts, Yvette Roke, Irene Niks, and Peter N. van Harten. 2023. Use of a mHealth mobile application to reduce stress in adults with autism: A pre-post pilot study of the stress autism Mate (SAM). *Advances in Neurodevelopmental Disorders* 7, 2 (June 2023), 268–276. DOI: <https://doi.org/10.1007/s41252-022-00304-3>
- [60] Valtteri Hongisto, Johanna Varjo, Henri Leppämäki, David Oliva, and Jukka Hyönä. 2016. Work performance in private office rooms: The effects of sound insulation and sound masking. *Building and Environment* 104 (August 2016), 263–274. DOI: <https://doi.org/10.1016/j.buildenv.2016.04.022>
- [61] Laura Hull, K. V. Petrides, Carrie Allison, Paula Smith, Simon Baron-Cohen, Meng Chuan Lai, and William Mandy. 2017. “Putting on My Best Normal”: Social camouflaging in adults with autism spectrum conditions. *Journal of Autism and Developmental Disorders* 47, 8 (2017), 2519–2534. DOI: <https://doi.org/10.1007/s10803-017-3166-5>
- [62] Nobuhiko Ikuta, Ryoichiro Iwanaga, Akiko Tokunaga, Hideyuki Nakane, Koji Tanaka, and Goro Tanaka. 2016. Effectiveness of earmuffs and noise-cancelling headphones for coping with hyper-reactivity to auditory stimuli in children with autism spectrum disorder: A preliminary study. *Hong Kong Journal of Occupational Therapy* 28, 1 (December 2016), 24–32. DOI: <https://doi.org/10.1016/j.hkjot.2016.09.001>
- [63] Muhammad Shoaib Jaliaawala and Rizwan Ahmed Khan. 2020. Can autism be catered with artificial intelligence-assisted intervention technology? A comprehensive survey. *Artificial Intelligence Review* 53, 2 (February 2020), 1039–1069. DOI: <https://doi.org/10.1007/s10462-019-09686-8>
- [64] Hanwoong Jung, Hexiang Ji, Alexey Pushchin, Maxim Ostapenko, Wenlong Niu, Ilya Palachev, Yutian Qu, Pavel Fedin, Yuri Gribov, Heewoo Nam, et al. 2023. Accelerating deep neural networks on mobile multicore NPUs. In *Proceedings of the 21st ACM/IEEE International Symposium on Code Generation and Optimization (CGO '23)*, ACM, 236–248. DOI: <https://doi.org/10.1145/3579990.3580015>
- [65] Niloofar Kalantari, Hui Zheng, Heidi J. Graff, Anna S. Evmenova, and Vivian Genaro Motti. 2021. Emotion regulation for neurodiversity through wearable technology. In *Proceedings of the 2021 9th International Conference on Affective Computing and Intelligent Interaction (ACII '21)*. Institute of Electrical and Electronics Engineers Inc. DOI: <https://doi.org/10.1109/ACII52823.2021.9597458>
- [66] Steven K. Kapp, Robyn Steward, Laura Crane, Daisy Elliott, Chris Elphick, Elizabeth Pellicano, and Ginny Russell. 2019. ‘People should be allowed to do what they like’: Autistic adults’ views and experiences of stimming. *Autism* 23, 7 (October 2019), 1782–1792. DOI: <https://doi.org/10.1177/1362361319829628>
- [67] Jessica M. Keith, Jeremy P. Jamieson, and Loisa Bennetto. 2019. The importance of adolescent self-report in autism spectrum disorder: Integration of questionnaire and autonomic measures. *Journal of Abnormal Child Psychology* 47, 4 (April 2019), 741–754. DOI: <https://doi.org/10.1007/s10802-018-0455-1>
- [68] Chelsea Kelling, Daniella Pitaro, and Jussi Rantala. 2016. Good vibes: The impact of haptic patterns on stress levels. In *Proceedings of the 20th International Academic Mindtrek Conference (AcademicMindtrek '16)*. ACM, 130–136. DOI: <https://doi.org/10.1145/2994310.2994368>
- [69] Kamran Khowaja, Dena Al-Thani, Asma Osman Hassan, Asadullah Shah, and Siti Salwah Salim. 2020. Mobile augmented reality app for children with autism spectrum disorder (ASD) to learn vocabulary (MARVoc): From the requirement gathering to its initial evaluation. In *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 424–437. DOI: https://doi.org/10.1007/978-3-030-50164-8_31
- [70] Kamran Khowaja, Bilikis Banire, Dena Al-Thani, Mohammed Tahri Sqalli, Aboubakr Agle, Asadullah Shah, and Siti Salwah Salim. 2020. Augmented reality for learning of children and adolescents with autism spectrum disorder (ASD): A systematic review. *IEEE Access* 8 (2020), 78779–78807. DOI: <https://doi.org/10.1109/ACCESS.2020.2986608>
- [71] Agata Kołakowska, Agnieszka Landowska, and Katarzyna Karpienko. 2017. Gyroscope-based game revealing progress of children with autism. In *ACM International Conference Proceeding Series*. ACM, 19–24. DOI: <https://doi.org/10.1145/3036290.3036324>
- [72] Yiannis Koumpourous and Theodoros Kafazis. 2019. Wearables and mobile technologies in autism spectrum disorder interventions: A systematic literature review. *Research in Autism Spectrum Disorders* 66 (2019). DOI: <https://doi.org/10.1016/j.rasd.2019.05.005>
- [73] Yiannis Koumpourous and Theodoros Kafazis. 2019. Wearables and mobile technologies in autism spectrum disorder interventions: A systematic literature review. *Research in Autism Spectrum Disorders* 66 (October 2019), 101405. DOI: <https://doi.org/10.1016/j.rasd.2019.05.005>
- [74] Aubrey J. Kumm, Marisa Viljoen, and Petrus J. de Vries. 2022. The digital divide in technologies for autism: Feasibility considerations for low- and middle-income countries. *Journal of Autism and Developmental Disorders* 52, 5 (May 2022), 2300–2313. DOI: <https://doi.org/10.1007/s10803-021-05084-8>

- [75] Toni R. Van Laarhoven, Jesse W. Johnson, Natalie R. Andzik, Lavisha Fernandes, Michael Ackerman, Maria Wheeler, Katherine Melody, Veronica Cornell, Gretta Ward, and Heather Kerfoot. 2021. Using Wearable Biosensor Technology in Behavioral Assessment for Individuals with Autism Spectrum Disorders and Intellectual Disabilities Who Experience Anxiety. DOI : <https://doi.org/10.1007/s41252-020-00191-6>/Published
- [76] Kate B. Lalonde, Brian R. MacNeill, Laura Wolfe Eversole, Steven P. Ragotzy, and Alan Poling. 2014. Increasing physical activity in young adults with autism spectrum disorders. *Research in Autism Spectrum Disorders* 8, 12 (December 2014), 1679–1684. DOI : <https://doi.org/10.1016/J.RASD.2014.09.001>
- [77] Fan Li, Huijie Chen, Xiaoyu Song, Qian Zhang, Youqi Li, and Yu Wang. 2017. CondioSense: High-quality context-aware service for audio sensing system via active sonar. *Personal and Ubiquitous Computing* 21, 1 (February 2017), 17–29. DOI : <https://doi.org/10.1007/s00779-016-0981-1>
- [78] Hailin Li, Bijun Shi, Xin Wang, Muqing Cao, Jiajie Chen, Siyu Liu, Xiaoling Zhan, Chengkai Jin, Zhaohuan Gui, Jin Jing, et al. 2022. Associations of emotional/behavioral problems with accelerometer-measured sedentary behavior, physical activity and step counts in children with autism spectrum disorder. *Frontiers in Public Health* 10 (October 2022). DOI : <https://doi.org/10.3389/fpubh.2022.981128>
- [79] Xuan Li, Huixin Zhong, Bin Zhang, and Jiaming Zhang. 2020. A general Chinese chatbot based on deep learning and its' application for children with ASD. *International Journal of Machine Learning* 10, 4 (2020), 519–526. DOI : <https://doi.org/10.18178/ijmlc.2020.10.4.967>
- [80] LinkedIn. 2024. LinkedIn. Retrieved April 6, 2022 from <https://uk.linkedin.com/>
- [81] P. Lipkin, J. Law, A. Marvin, E. Rubenstein, and J. Toroney. 2016. Auditory hypersensitivity issues in children with autism spectrum disorders (ASD): Characteristics and burden. *Annals of Neurology* 80 (2016), S392. DOI : <https://doi.org/10.1002/pon.4272>
- [82] Luxottica Group S.p.A. 2024. Ray-Ban | Meta Smart Glasses 2024 | Ray-Ban® GB. Retrieved June 26, 2024 from <https://www.ray-ban.com/uk/ray-ban-meta-smart-glasses>
- [83] Tommy Mackay, June Ward, Chris Cole, Mary Mccutcheon, and Ken Aitken. 2013. Good autism practice the Scottish perspective. *Good Autism Practice* October 14, Suppl 1 (2013).
- [84] Matthew J. Maenner, Zachary Warren, Ashley Robinson Williams, Esther Amoakohene, Amanda V. Bakian, Deborah A. Bilder, Maureen S. Durkin, Robert T. Fitzgerald, Sarah M. Furnier, Michelle M. Hughes, et al. 2023. Prevalence and characteristics of autism spectrum disorder among children aged 8 years — Autism and developmental disabilities monitoring network, 11 Sites, United States, 2020. *MMWR. Surveillance Summaries* 72, 2 (March 2023), 1–14. DOI : <https://doi.org/10.15585/mmwr.ss7202a1>
- [85] Magic Leap Inc. 2024. Magic Leap 2 | Magic Leap. Retrieved June 27, 2024 from <https://www.magicleap.com/magic-leap-2>
- [86] Janessa Manning, Joseph Billian, Jill Matson, Colleen Allen, and Neelkamal Soares. 2021. Perceptions of families of individuals with autism spectrum disorder during the COVID-19 crisis. *Journal of Autism and Developmental Disorders* 51, 8 (August 2021), 2920–2928. DOI : <https://doi.org/10.1007/s10803-020-04760-5>
- [87] Nicholas Mariette. 2013. *Human factors research in audio augmented reality*. In *Human Factors in Augmented Reality Environments*. Springer, New York, 11–32. DOI : https://doi.org/10.1007/978-1-4614-4205-9_2
- [88] Morag Maskey, Jessica Lowry, Jacqui Rodgers, Helen McConachie, and Jeremy R. Parr. 2014. Reducing specific phobia/fear in young people with autism spectrum disorders (ASDs) through a virtual reality environment intervention. *PLoS One* 9, 7 (July 2014), e100374. DOI : <https://doi.org/10.1371/journal.pone.0100374>
- [89] Laura Mattys, Ilse Noens, Kris Evers, and Dieter Baeyens. 2018. “Hold Me Tight So I Can Go It Alone”: Developmental themes for young adults with autism spectrum disorder. *Qualitative Health Research* 28, 2 (January 2018), 321–333. DOI : <https://doi.org/10.1177/1049732317730329>
- [90] Laura Mattys, Ilse Noens, Kris Evers, and Dieter Baeyens. 2018. “Hold Me Tight So I Can Go It Alone”: Developmental themes for young adults with autism spectrum disorder. *Qualitative Health Research* 28, 2 (January 2018), 321–333. DOI : <https://doi.org/10.1177/1049732317730329>
- [91] John Joseph McGowan and Iain Peter McGregor. 2023. Investigation into stress triggers in autistic adults for the development of technological self-interventions. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '23)*. ACM, Inc. DOI : <https://doi.org/10.1145/3597638.3608392>
- [92] MeloAudio. 2023. Enhanced Situational Awareness Headphones: Redefining the Audio Experi. Retrieved June 26, 2024 from <https://meloaudio.com/it/blogs/blog/enhanced-situational-awareness-headphones>
- [93] Meta. 2024. Meta Quest VR Headsets, Accessories and Equipment | Meta Quest | Meta Store. Retrieved June 27, 2024 from https://www.meta.com/gb/quest/?utm_source=www.oculus.com&utm_medium=oculusredirect
- [94] Meta. 2024. Meta's Progress in Augmented and Virtual Reality | Meta. Retrieved June 26, 2024 from <https://about.fb.com/news/2022/12/metasp-progress-in-augmented-and-virtual-reality/>
- [95] Microsoft. 2024. Microsoft Teams. Retrieved February 16, 2024 from <https://www.microsoft.com/en-gb/microsoft-teams/log-in>

- [96] Microsoft. 2024. HoloLens 2—Overview, Features, and Specs | Microsoft HoloLens. Retrieved June 27, 2024 from <https://www.microsoft.com/en-us/hololens/hardware#document-experiences>
- [97] MoldStud. 2024. Implementing Gesture Controls in Android App Development | MoldStud. Retrieved June 24, 2024 from <https://moldstud.com/articles/p-implementing-gesture-controls-in-android-app-development>
- [98] Sun Jae Moon, Jinseub Hwang, Harrison Scott Hill, Ryan Kervin, Kirstin Brown Birtwell, John Torous, Christopher J. McDougale, and Jung Won Kim. 2020. Mobile device applications and treatment of autism spectrum disorder: A systematic review and meta-analysis of effectiveness. *Archives of Disease in Childhood* 105, 5 (May 2020), 458–462. DOI: <https://doi.org/10.1136/archdischild-2019-318258>
- [99] Laurent Mottron, Michelle Dawson, Isabelle Soulières, Benedicte Hubert, and Jake Burack. 2006. Enhanced perceptual functioning in autism: An update, and eight principles of autistic perception. *Journal of Autism and Developmental Disorders* 36 (2006), 27–43. DOI: <https://doi.org/10.1007/s10803-005-0040-7>
- [100] Elias Mpofu, Rebecca Cagle, Chung-Yi Chiu, Qiwei Li, and Linda Holloway. 2021. Digital tools applications to occupational health and safety for people with autism. In *Agile Coping in the Digital Workplace*. Springer International Publishing, Cham, 147–165. DOI: https://doi.org/10.1007/978-3-030-70228-1_8
- [101] Brenda Smith Myles, Deborah E. Griswold, and Richard L. Simpson. 2001. A synthesis of studies on the intellectual, academic, social/emotional and sensory characteristics of children and youth with Asperger syndrome. Retrieved from <https://www.researchgate.net/publication/279545129>
- [102] Emily Neuhaus, Theodore P. Beauchaine, Raphael A. Bernier, and Sara J. Webb. 2018. Child and family characteristics moderate agreement between caregiver and clinician report of autism symptoms. *Autism Research* 11, 3 (March 2018), 476–487. DOI: <https://doi.org/10.1002/aur.1907>
- [103] Nigel Newbutt, Connie Sung, Hung Jen Kuo, and Michael J. Leahy. 2017. The acceptance, challenges, and future applications of wearable technology and virtual reality to support people with autism spectrum disorders. In *Intelligent Systems Reference Library*. Springer Science and Business Media Deutschland GmbH, 221–241. DOI: https://doi.org/10.1007/978-3-319-49879-9_11
- [104] Nuheara. 2021. Hear Through Earbuds: Many Names with a Common Purpose | Nuheara. Retrieved June 24, 2024 from <https://www.nuheara.com/news/hear-through-earbuds/>
- [105] Samuel L. Odom, Julie L. Thompson, Susan Hedges, Brian A. Boyd, Jessica R. Dykstra, Michelle A. Duda, Kathrine L. Szidon, Leann E. Smith, and Aimee Bord. 2015. Technology-aided interventions and instruction for adolescents with autism spectrum disorder. *Journal of Autism and Developmental Disorders* 45, 12 (2015), 3805–3819. DOI: <https://doi.org/10.1007/s10803-014-2320-6>
- [106] Kathryn N. Oriel, Jennifer Wood Kanupka, Adam T. Fuehrer, Kayla M. Klumpp, Katelyn N. Stoltz, Daniel W. Willey, and Michael L. Decvalcante. 2018. The impact of a rock climbing program for adolescents with autism spectrum disorder: A pilot study. *International Journal of Kinesiology in Higher Education* 2, 4 (October 2018), 113–126. DOI: <https://doi.org/10.1080/24711616.2018.1425601>
- [107] Ketan R. Parmar, Catherine S. Porter, Christine M. Dickinson, James Pelham, Peter Baimbridge, and Emma Gowen. 2021. Visual sensory experiences from the viewpoint of autistic adults. *Frontiers in Psychology* 12 (June 2021). DOI: <https://doi.org/10.3389/fpsyg.2021.633037>
- [108] Isabel Paula-Pérez. 2013. Differential diagnosis between obsessive compulsive disorder and restrictive and repetitive behavioural patterns, activities and interests in autism spectrum disorders. *Revista de Psiquiatría y Salud Mental (English Edition)* 6, 4 (October 2013), 178–186. DOI: <https://doi.org/10.1016/j.rpsmen.2012.07.006>
- [109] Susan Peppé, Joanne McCann, Fiona Gibbon, Anne O'Hare, and Marion Rutherford. 2007. Receptive and expressive prosodic ability in children with high-functioning autism. *Journal of Speech, Language, and Hearing Research* 50, 4 (August 2007), 1015–1028. DOI: [https://doi.org/10.1044/1092-4388\(2007/071\)](https://doi.org/10.1044/1092-4388(2007/071))
- [110] A. Pérusse-Lambert, M. Anastassova, M. Boukallel, M. Chetouani, and O. Grynzspan. 2016. Engaging children with autism in interaction using haptic and tactile interfaces. In *ICMI 2016-Proceedings of the 18th ACM International Conference on Multimodal Interaction*, 3–6.
- [111] Alix Pérusse-Lambert, M. Anastassova, M. Boukallel, M. Chetouani, and O. Grynzspan. 2016. Engaging children with autism in interaction using haptic and tactile interfaces. In *Proceedings of the 18th ACM International Conference on Multimodal Interaction (ICMI '16)*, 3–6. Retrieved from <https://ido2016.sciencesconf.org/123292/document>
- [112] Beth Pfeiffer, Moya Kinnealey, Carol Reed, and Georgiana Herzberg. 2005. Sensory modulation and affective disorders in children and adolescents with Asperger's disorder. *American Journal of Occupational Therapy* 59, 3 (2005), 335–345. DOI: <https://doi.org/10.5014/ajot.59.3.335>
- [113] Beth Pfeiffer, Leah Stein Duker, Ann Marie Murphy, and Chengshi Shui. 2019. Effectiveness of noise-attenuating headphones on physiological responses for children with autism spectrum disorders. *Frontiers in Integrative Neuroscience* 13 (November 2019). DOI: <https://doi.org/10.3389/fnint.2019.00065>
- [114] Proton AG. 2024. What Is GDPR, the EU's New Data Protection Law? - GDPR.eu. Retrieved June 27, 2024 from <https://gdpr.eu/what-is-gdpr/>

- [115] Cynthia Putnam and Lorna Chong. 2008. Software and technologies designed for people with autism: what do users want? In *Proceedings of the 10th International ACM SIGACCESS Conference on Computers and Accessibility*, 3–10. DOI: <https://doi.org/10.1145/1414471.1414475>
- [116] QSR International Pty Ltd. 2017. NVivo Product Range | QSR International. Retrieved November 28, 2025 from <http://lumivero.com/products/nvivo/>
- [117] Shantha Radhakrishna. 2010. Application of integrated yoga therapy to increase imitation skills in children with autism spectrum disorder. *International Journal of Yoga* 3, 1 (2010), 26. DOI: <https://doi.org/10.4103/0973-6131.66775>
- [118] Forough Raffi, Naimeh Seyedfatemi, and Hamideh Elahi Asgarabad. 2024. The life of mothers of children with Autism: A grounded theory study. *Revista Latinoamericana de Hipertension* 19, 4 (2024), 154. DOI: <https://doi.org/10.5281/zenodo.11260914>
- [119] Daniel Ray, Tim Collins, Sandra Woolley, and Prasad Ponnappalli. 2023. A review of wearable multi-wavelength photoplethysmography. *IEEE Reviews in Biomedical Engineering* 16 (2023), 136–151. DOI: <https://doi.org/10.1109/RBME.2021.3121476>
- [120] Red Hat Inc. 2024. How to Customize Your Voice Assistant with the Voice of Your Choice | Opensource.com. Retrieved June 26, 2024 from <https://opensource.com/article/21/1/customize-voice-assistant>
- [121] Ikram Ur Rehman, Drishty Sobnath, Moustafa M. Nasralla, Maria Winnett, Aamir Anwar, Waqar Asif, and Hafiz Husnain Raza Sherazi. 2021. Features of mobile apps for people with autism in a post COVID-19 scenario: Current status and recommendations for apps using AI. *Diagnostics* 11, 10 (October 2021), 1923. DOI: <https://doi.org/10.3390/diagnostics11101923>
- [122] A E Robertson and Simmons. 2015. The Sensory Experiences of Adults with Autism Spectrum Disorder: A Qualitative Analysis. Retrieved from <http://eprints.gla.ac.uk/106809/>
- [123] Ashley E. Robertson and David R. Simmons. 2015. The sensory experiences of adults with autism spectrum disorder: A qualitative analysis. *Perception* 44, 5 (2015), 569–586. DOI: <https://doi.org/10.1068/p7833>
- [124] Ashley E. Robertson and David R. Simmons. 2015. The sensory experiences of adults with autism spectrum disorder: A qualitative analysis. *Perception* 44, 5 (May 2015), 569–586. DOI: <https://doi.org/10.1068/p7833>
- [125] Monica R. Romero, Estela Macas, Ivana Harari, and Javier Diaz. 2020. *Applied Technologies*. Springer International Publishing, Cham. DOI: <https://doi.org/10.1007/978-3-030-42520-3>
- [126] Mohammed F. Safi, Badriya Al Sadrani, and Ashraf Mustafa. 2023. Virtual voice assistant applications improved expressive verbal abilities and social interactions in children with autism spectrum disorder: a single-subject experimental study. *International Journal of Development Disabilities* 69, 4 (2023), 555–567. DOI: <https://doi.org/10.1080/20473869.2021.1977596>
- [127] Hillol Sarker, Allison Tam, Morgan Foreman, Nicholas Fay, Murtaza Dhuliawala, and Amar Das. 2018. Detection of Stereotypical Motor Movements in Autism using a Smartwatch-based System. In *AMIA Annual Symposium Proceedings*, 952.
- [128] Manuela Schuetze, Christiane S. Rohr, Deborah Dewey, Adam McCrimmon, and Signe Bray. 2017. Reinforcement learning in autism spectrum disorder. *Frontiers in Psychology* 8 (2017). DOI: <https://doi.org/10.3389/fpsyg.2017.02035>
- [129] Lorenzo Seidenari, Claudio Baecchi, Tiberio Uricchio, Andrea Ferracani, Marco Bertini, and Alberto Del Bimbo. 2017. Deep artwork detection and retrieval for automatic context-aware audio guides. *ACM Transactions on Multimedia Computing, Communications and Applications* 13, 3s (June 2017). DOI: <https://doi.org/10.1145/3092832>
- [130] Seizeyourautism. 2022. Seize Your Autism — Guide to Meditation with ASD. Retrieved April 8, 2022 from <https://seizeyourautism.tumblr.com/post/139625059069/guide-for-meditation-with-asd>
- [131] Randye J. Semple. 2019. Review: Yoga and mindfulness for youth with autism spectrum disorder: Review of the current evidence. *Child and Adolescent Mental Health* 24, 1 (February 2019), 12–18. DOI: <https://doi.org/10.1111/camh.12295>
- [132] Sensory Direct. 2024. Fidget Toys & Fiddle Toys for Autism & ADHD. Sensory Direct. Retrieved March 21, 2024 from <https://www.sensorydirect.com/behaviour/concentration/fidget-toys>
- [133] Fatin Amira Shahrudin, Ahmad Aidil Arafat Dzulkarnain, Ayu Madiha Hanafi, Fatin Nabilah Jamal, Nadzirah Ahmad Basri, Shahrul Na’Im Sidek, Hazlina Md Yusof, and Madihah Khalid. 2022. Music and sound-based intervention in autism spectrum disorder: A scoping review. *Psychiatry Investig* 19, 8 (August 2022), 626–636. DOI: <https://doi.org/10.30773/pi.2021.0382>
- [134] Nirbhay N. Singh, Giulio E. Lancioni, Ramasamy Manikam, Alan S.W. Winton, Ashvind N.A. Singh, Judy Singh, and Angela D.A. Singh. 2011. A mindfulness-based strategy for self-management of aggressive behavior in adolescents with autism. *Research in Autism Spectrum Disorders* 5, 3 (July 2011), 1153–1158. DOI: <https://doi.org/10.1016/j.rasd.2010.12.012>
- [135] Paige M. Siper, Alexander Kolevzon, A. Ting Wang, Joseph D. Buxbaum, and Teresa Tavassoli. 2017. A clinician-administered observation and corresponding caregiver interview capturing DSM-5 sensory reactivity symptoms in children with ASD. *Autism Research* 10, 6 (June 2017), 1133–1140. DOI: <https://doi.org/10.1002/aur.1750>

- [136] Richard S. Smith and Jonathan Sharp. 2013. Fascination and isolation: A grounded theory exploration of unusual sensory experiences in adults with Asperger syndrome. *Journal of Autism and Developmental Disorders* 43, 4 (April 2013), 891–910. DOI: <https://doi.org/10.1007/s10803-012-1633-6>
- [137] Debbie Spain, David Mason, Simone J. Capp, Laura Stoppelbein, Susan W. White, and Francesca Happé. 2021. “This may be a really good opportunity to make the world a more autism friendly place”: Professionals’ perspectives on the effects of COVID-19 on autistic individuals. *Research in Autism Spectrum Disorders* 83 (May 2021). DOI: <https://doi.org/10.1016/j.rasd.2021.101747>
- [138] Katharina Spiel, Christopher Frauenberger, and Geraldine Fitzpatrick. 2017. Experiences of autistic children with technologies. *International Journal of Child-Computer Interaction* 11 (2017), 50–61. DOI: <https://doi.org/10.1016/j.ijcci.2016.10.007>
- [139] Jennifer Stephenson and Lisa Limbrick. 2015. A review of the use of touch-screen mobile devices by people with developmental disabilities. *Journal of Autism and Developmental Disorders* 45, 12 (2015). DOI: <https://doi.org/10.1007/s10803-013-1878-8>
- [140] Lillian N. Stiegler and Rebecca Davis. 2010. Understanding sound sensitivity in individuals with autism spectrum disorders. *Focus on Autism and Other Developmental Disabilities* 25, 2 (2010), 67–75. DOI: <https://doi.org/10.1177/1088357610364530>
- [141] Dorothy Strickland, Lee M. Marcus, Gary B. Mesibov, and Kerry Hogan. 1996. Brief report: Two case studies using virtual reality as a learning tool for autistic children. *Journal of Autism and Developmental Disorders* 26, 6 (1996), 651–659. DOI: <https://doi.org/10.1007/BF02172354>
- [142] Xia Su, Eunye Koh, and Chang Xiao. 2024. SonifyAR: Context-aware sound effect generation in augmented reality. *ACM*, 1–7. DOI: <https://doi.org/10.1145/3613905.3650927>
- [143] Sharifah Syed Abdullah, Dayang Awang Rambli, Suziah Sulaiman, Emad Alyan, Frederic Merienne, and Nadia Diyana Mohd Muhaiddin. 2021. The impact of virtual nature therapy on stress responses: A systematic qualitative review. *Forests* 12, 12 (December 2021), 1776. DOI: <https://doi.org/10.3390/f12121776>
- [144] Ziyang Tang, Jin Guo, Sheng Miao, Subrata Acharya, and Jinjuan Heidi Feng. 2016. Ambient intelligence based context-aware assistive system to improve independence for people with autism spectrum disorder. In *Proceedings of the 2016 49th Hawaii International Conference on System Sciences (HICSS)*. IEEE, 3339–3348. DOI: <https://doi.org/10.1109/HICSS.2016.417>
- [145] Zeina Thabet, Hurmat Ansari, Sara Albashtawi, Nur Siyam, and Sherief Abdallah. 2023. Using deep learning and native mobile app to assist autistic students’ educational experience. In *Proceedings of the 15th International Conference on Computer Supported Education*. SCITEPRESS - Science and Technology Publications, 96–103. DOI: <https://doi.org/10.5220/0011748800003470>
- [146] The Autistic Innovator. 2024. Stim Toys for Autistic Adults – The Autistic Innovator. Retrieved May 10, 2024 from <https://shop.autisticinnovator.com/en-gb/collections/stim-toys>
- [147] Thruday Ltd. 2024. Visual Daily Planning App for Autism, ADHD, ADD. Retrieved June 27, 2024 from <https://thruday.com/>
- [148] Twitter. 2022. Twitter. It’s What’s Happening/Twitter. Twitter.com. Retrieved April 6, 2022 from <https://twitter.com/>
- [149] Jenny A. Vlachou and Athanasios S. Drigas. 2017. Mobile technology for students & adults with Autistic Spectrum Disorders (ASD). *International Journal of Interactive Mobile Technologies* 11, 1s (2017), 4–17. DOI: <https://doi.org/10.3991/ijim.v11i1.5922>
- [150] Helmut R. Wagner, Barney G. Glaser, and Anselm L. Strauss. 1968. The discovery of grounded theory: Strategies for qualitative research. *Social Forces* 46, 4 (June 1968), 555. DOI: <https://doi.org/10.2307/2575405>
- [151] Lina J. Wali and Filippo Sanfilippo. 2019. A review of the state-of-the-art of assistive technology for people with ASD in the workplace and in everyday life. In *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*. Springer Verlag, 520–532. DOI: https://doi.org/10.1007/978-3-030-29374-1_42
- [152] Manhua Wang and Myounghoon Jeon. 2023. Assistive technology for adults on the autism spectrum: A systematic survey. *International Journal of Human-Computer Interaction* 40 (2023), 2433–2452. DOI: <https://doi.org/10.1080/10447318.2022.2163568>
- [153] Lisa Weston, Joanne Hodgekins, and Peter E. Langdon. 2016. Effectiveness of cognitive behavioural therapy with people who have autistic spectrum disorders: A systematic review and meta-analysis. *Clinical Psychology Review* 49 (November 2016), 41–54. DOI: <https://doi.org/10.1016/j.cpr.2016.08.001>
- [154] Susan W. White, Rebecca Elias, Carlos E. Salinas, Nicole Capriola, Caitlin M. Conner, Susan B. Asselin, Yasuo Miyazaki, Carla A. Mazefsky, Patricia Howlin, and Elizabeth E. Getzel. 2016. Students with autism spectrum disorder in college: Results from a preliminary mixed methods needs analysis. *Research in Developmental Disabilities* 56 (September 2016), 29–40. DOI: <https://doi.org/10.1016/j.ridd.2016.05.010>

- [155] Julia Woodward and Jaime Ruiz. 2023. Analytic review of using augmented reality for situational awareness. *IEEE Transactions on Visualization and Computer Graphics* 29, 4 (April 2023), 2166–2183. DOI: <https://doi.org/10.1109/TVCG.2022.3141585>
- [156] Sandra I. Woolley, Tim Collins, James Mitchell, and David Fredericks. 2019. Investigation of wearable health tracker version updates. *BMJ Health & Care Informatics* 26, 1 (October 2019). DOI: <https://doi.org/10.1136/bmjhci-2019-100083>
- [157] Mengting Xia, Nan Chen, Yuemin Tang, and Zhigang Zhu. 2020. ParaShop: A mobile AR app in assisting people with ASD in shopping. In *Proceedings of the 2020 IEEE MIT Undergraduate Research Technology Conference (URTC)*. IEEE, 1–4. DOI: <https://doi.org/10.1109/URTC51696.2020.9668867>
- [158] Ozgur Yurur, Chi Harold Liu, Zhengguo Sheng, Victor C. M. Leung, Wilfrido Moreno, and Kin K. Leung. 2016. Context-awareness for mobile sensing: A survey and future directions. *IEEE Communications Surveys and Tutorials* 18, 1 (January 2016), 68–93. DOI: <https://doi.org/10.1109/COMST.2014.2381246>
- [159] Camellia Zakaria and Richard C. Davis. 2016. Demo. In *Proceedings of the 14th Annual International Conference on Mobile Systems, Applications, and Services Companion*. ACM, New York, NY, 127–127. DOI: <https://doi.org/10.1145/2938559.2938575>
- [160] Camellia Zakaria, Richard C. Davis, and Zachary Walker. 2016. Seeking independent management of problem behavior. In *Proceedings of the 15th International Conference on Interaction Design and Children*. ACM, New York, NY, 196–205. DOI: <https://doi.org/10.1145/2930674.2930693>
- [161] Catherine Zenko. 2014. Practical solutions for executive function challenges created by the unique learning styles of students with autism spectrum disorder (ASD). *Perspectives on School-Based Issues* 15, 4 (December 2014), 141–150. DOI: <https://doi.org/10.1044/sbi15.4.141>
- [162] Keshnerim Ruben. 2024. Autism and Sound Sensitivity: Silencing the Chaos. Retrieved June 23, 2024 from <https://www.supportivecareaba.com/aba-therapy/autism-and-sound-sensitivity>

Received 28 June 2024; revised 14 October 2025; accepted 31 October 2025